



US012401752B2

(12) **United States Patent**
Ogura

(10) **Patent No.:** **US 12,401,752 B2**

(45) **Date of Patent:** ***Aug. 26, 2025**

(54) **IMAGE PROCESSING APPARATUS**

(71) Applicant: **Brother Kogyo Kabushiki Kaisha,**
Nagoya (JP)

(72) Inventor: **Sho Ogura,** Nagoya (JP)

(73) Assignee: **Brother Kogyo Kabushiki Kaisha,**
Nagoya (JP)

(*) Notice: Subject to any disclaimer, the term of this
patent is extended or adjusted under 35
U.S.C. 154(b) by 0 days.

This patent is subject to a terminal dis-
claimer.

(21) Appl. No.: **18/435,783**

(22) Filed: **Feb. 7, 2024**

(65) **Prior Publication Data**

US 2024/0179252 A1 May 30, 2024

Related U.S. Application Data

(63) Continuation of application No. 17/385,352, filed on
Jul. 26, 2021, now Pat. No. 11,943,401, which is a
continuation of application No. 16/581,072, filed on
Sep. 24, 2019, now Pat. No. 11,082,572.

(30) **Foreign Application Priority Data**

Sep. 28, 2018 (JP) 2018-184739

(51) **Int. Cl.**
H04N 1/00 (2006.01)

(52) **U.S. Cl.**
CPC **H04N 1/00514** (2013.01); **H04N 1/00432**
(2013.01); **H04N 2201/0094** (2013.01)

(58) **Field of Classification Search**

None

See application file for complete search history.

(56) **References Cited**

U.S. PATENT DOCUMENTS

2006/0206834 A1 9/2006 Fisher et al.
2010/0105365 A1* 4/2010 Kakiwaki H04M 1/724
455/566

(Continued)

FOREIGN PATENT DOCUMENTS

JP 2008-533593 A 8/2008
JP 2011-123824 A 6/2011

(Continued)

OTHER PUBLICATIONS

Apr. 2, 2024—(JP) Notice of Reasons for Refusal—JP App 2023-
130048, Eng Tran.

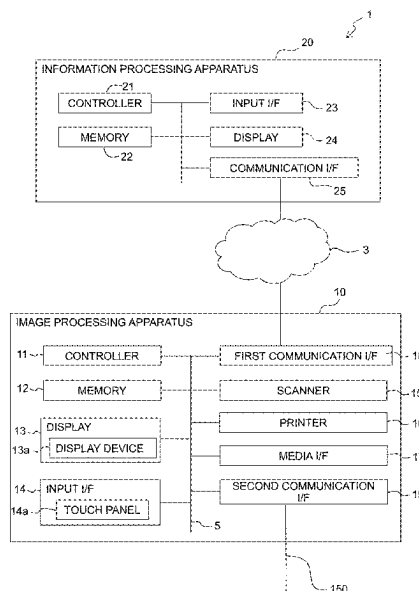
Primary Examiner — Frantz Bataille

(74) *Attorney, Agent, or Firm* — Banner & Witcoff, Ltd.

(57) **ABSTRACT**

An image processing apparatus includes: a printer; a user interface; and a controller. In a case that the controller receives an input of authentication information of a user and that the user has logged in to the image processing apparatus based on the authentication information, the controller causes the user interface to display one screen associated with the user logged in to the image processing apparatus. The one screen is one of a plurality of ordered screens and is neither a top screen nor a last screen of the plurality of ordered screens. The one screen is displayed on the user interface in a state of being switchable to a previous screen previous to the one screen and a next screen next to the one screen.

15 Claims, 19 Drawing Sheets



(56)

References Cited

U.S. PATENT DOCUMENTS

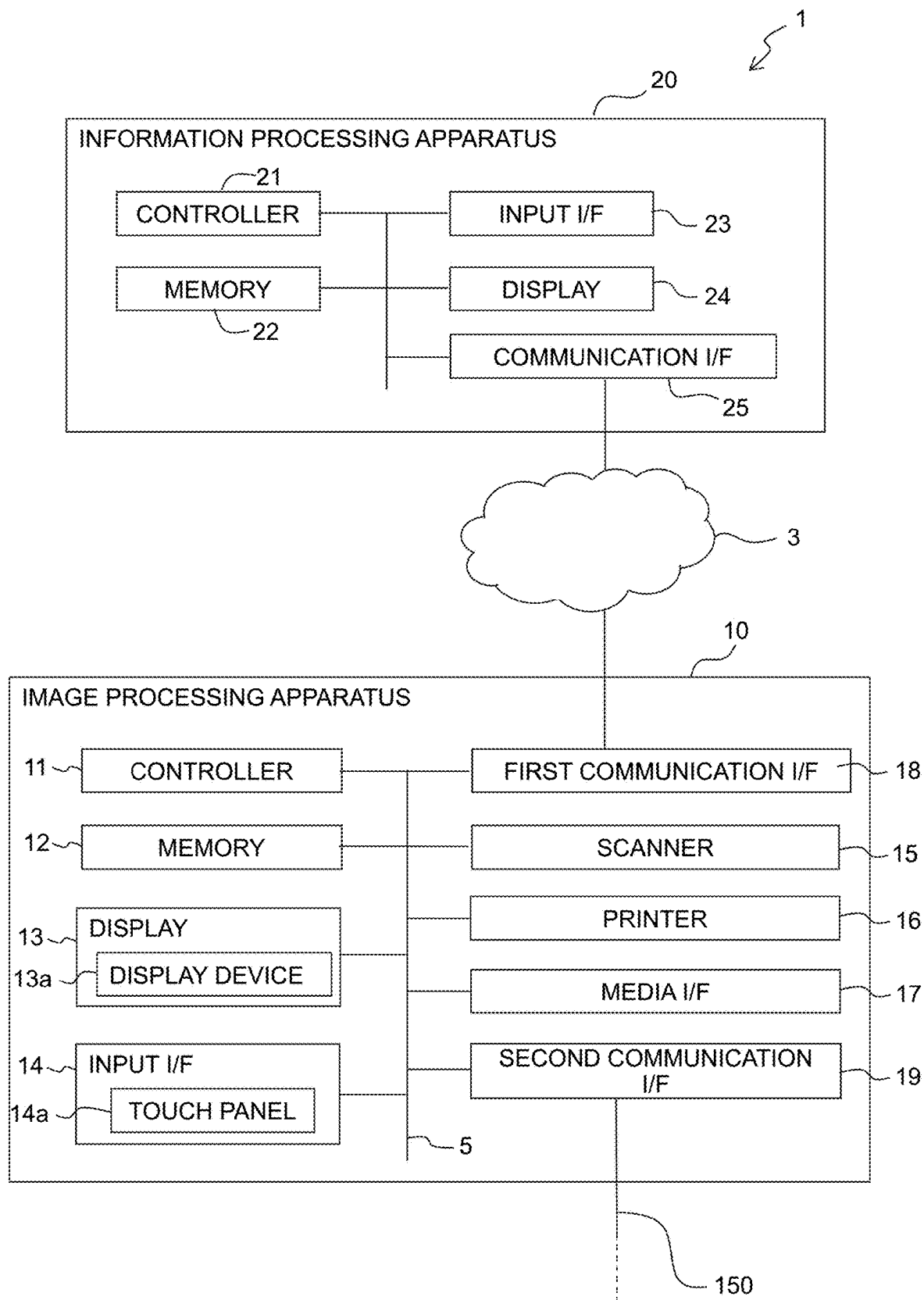
2012/0099130	A1 *	4/2012	Tohki	G03G 15/502
				358/1.13
2014/0101138	A1 *	4/2014	Sugita	G06F 16/248
				715/777
2014/0300920	A1	10/2014	Hayashi	

FOREIGN PATENT DOCUMENTS

JP	2016-066258	A	4/2016
JP	2016-071680	A	5/2016
JP	2018-112960	A	7/2018
JP	2018-152870	A	9/2018

* cited by examiner

FIG. 1



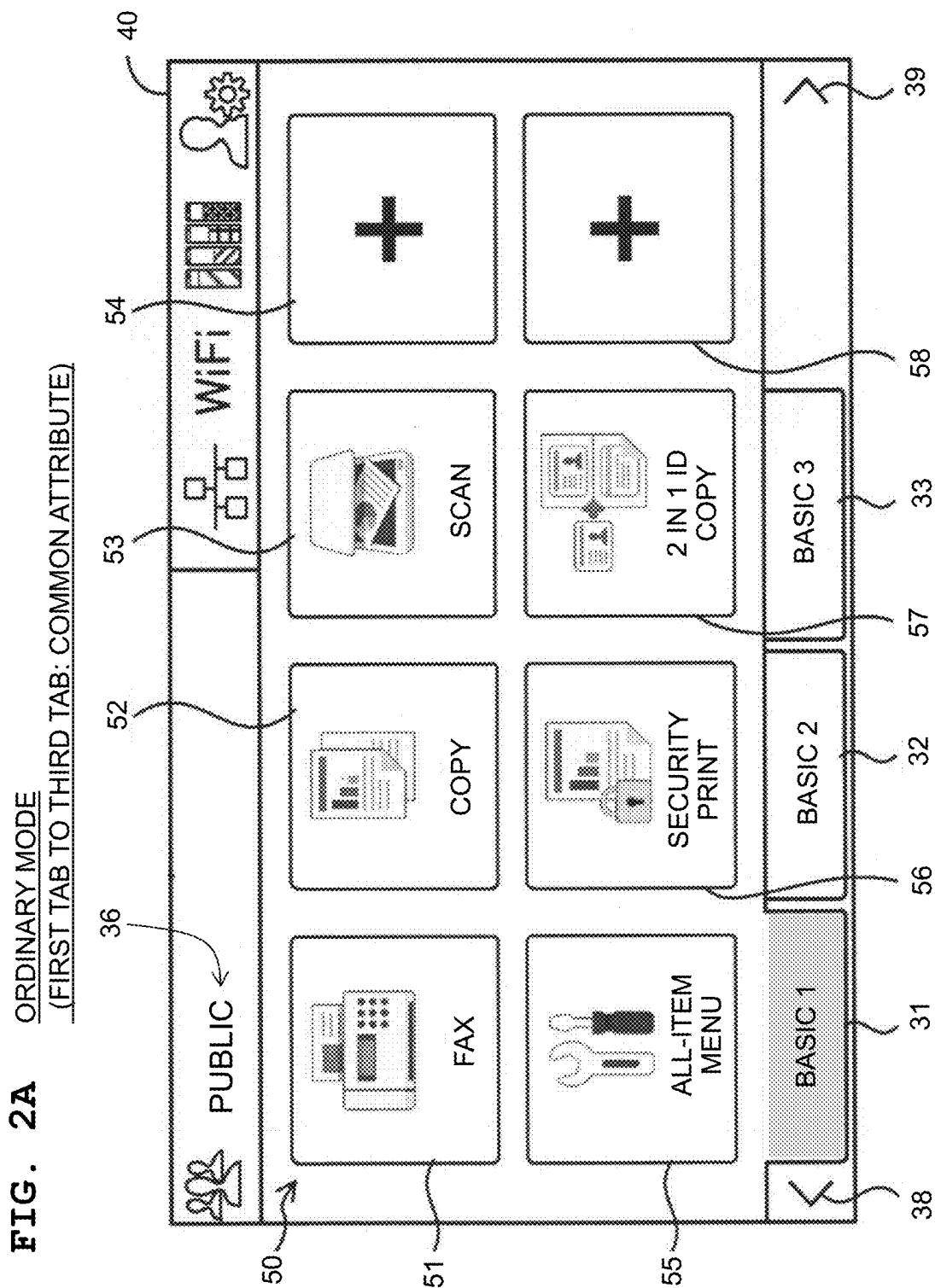


FIG. 2B

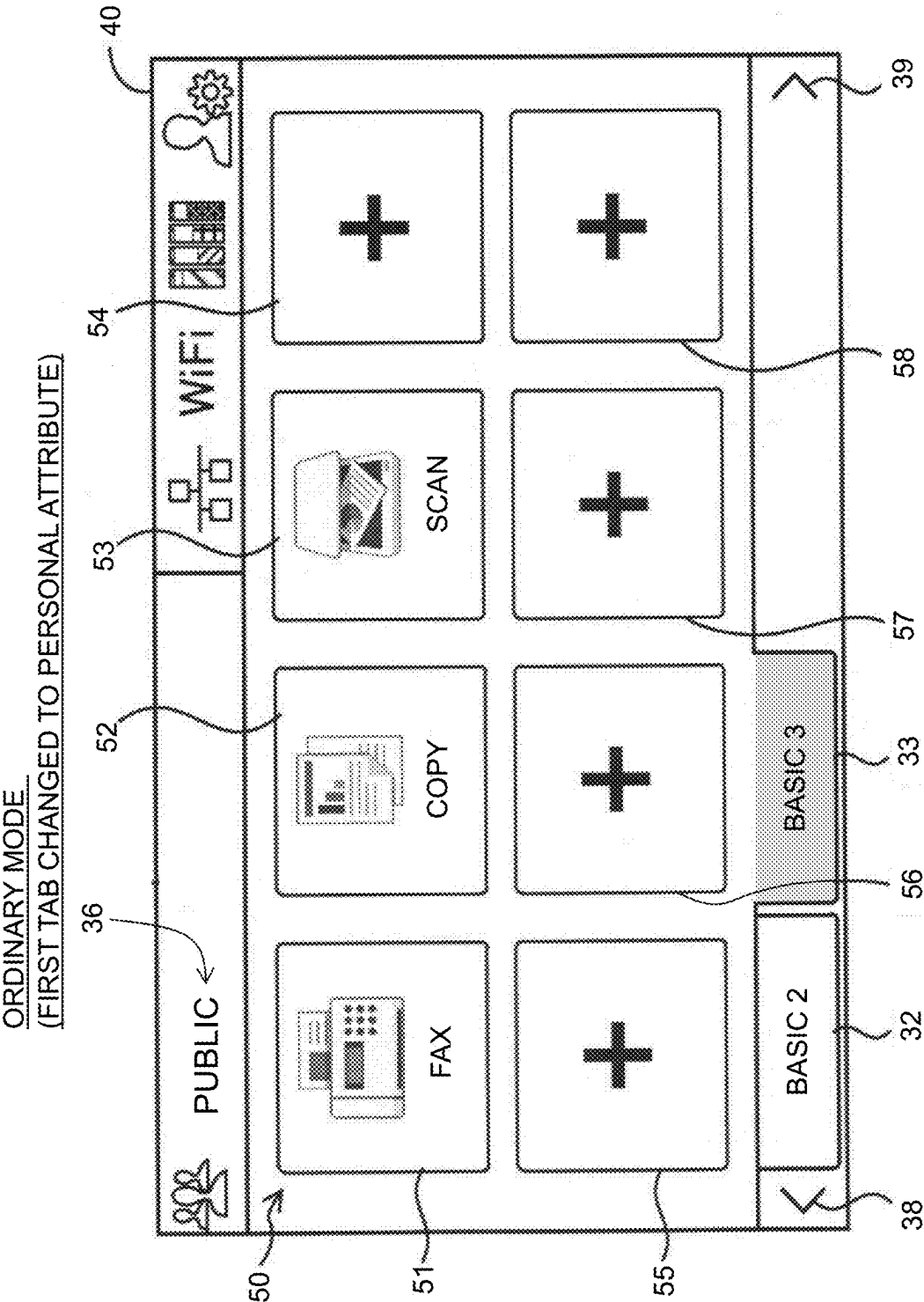


FIG. 3A

MAIN HOME SCREEN TAB SETTING TABLE

MAIN HOME SCREEN TAB	FIRST TAB
----------------------	-----------

FIG. 3B

TAB SETTING TABLE

INDEX	TAB ATTRIBUTE	TAB NAME	ICON NUMBER
FIRST TAB	COMMON	BASIC 1	6
SECOND TAB	COMMON	BASIC 2	0
THIRD TAB	COMMON	BASIC 3	3
FOURTH TAB	PERSONAL	CUSTOM 1	1
FIFTH TAB	PERSONAL	CUSTOM 2	1
SIXTH TAB	PERSONAL	CUSTOM 3	1
SEVENTH TAB	PERSONAL	CUSTOM 4	1
EIGHTH TAB	PERSONAL	CUSTOM 5	1
⋮	⋮	⋮	⋮
N-TH TAB	PERSONAL	xxx	

FIG. 3C

USER LIST

USER NAME	INDIVIDUAL REGISTRATION TAB	INDIVIDUAL MAIN HOME SCREEN TAB
USER A	FOURTH TAB	FOURTH TAB
USER B	FIFTH TAB	FIRST TAB
USER C	SECOND TAB	SECOND TAB
USER D	OFF	OFF
⋮	⋮	⋮

FIG. 3D

INDIVIDUAL MAIN HOME SCREEN TAB SETTING
TABLE OF INDIVIDUAL REGISTRATION TAB

SET INDIVIDUAL REGISTRATION TAB AS INDIVIDUAL MAIN HOME SCREEN TAB	ON
---	----

FIG. 4

7

30

TAB ATTRIBUTE SETTING SCREEN

71 → FIRST TAB ☒ COMMON ☐ PERSONAL ★

72 → SECOND TAB ☒ COMMON ☐ PERSONAL

73 → THIRD TAB ☒ COMMON ☐ PERSONAL

74 → FOURTH TAB ☐ COMMON ☒ PERSONAL

75 → FIFTH TAB ☐ COMMON ☒ PERSONAL

76 → SIXTH TAB ☐ COMMON ☒ PERSONAL

77 → SEVENTH TAB ☐ COMMON ☒ PERSONAL

78 → EIGHTH TAB ☐ COMMON ☒ PERSONAL

FIG. 5

35

MAIN HOME SCREEN SETTING

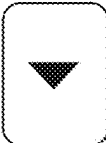
35a	FIRST TAB	BASIC 1	
35b	SECOND TAB	BASIC 2	
35c	THIRD TAB	BASIC 3	
35d	FOURTH TAB	CUSTOM 1	
35e	FIFTH TAB	CUSTOM 2	

FIG. 6A

INDIVIDUAL MODE
(SET INDIVIDUAL REGISTRATION TAB AS INDIVIDUAL MAIN
HOME SCREEN TAB: ON)

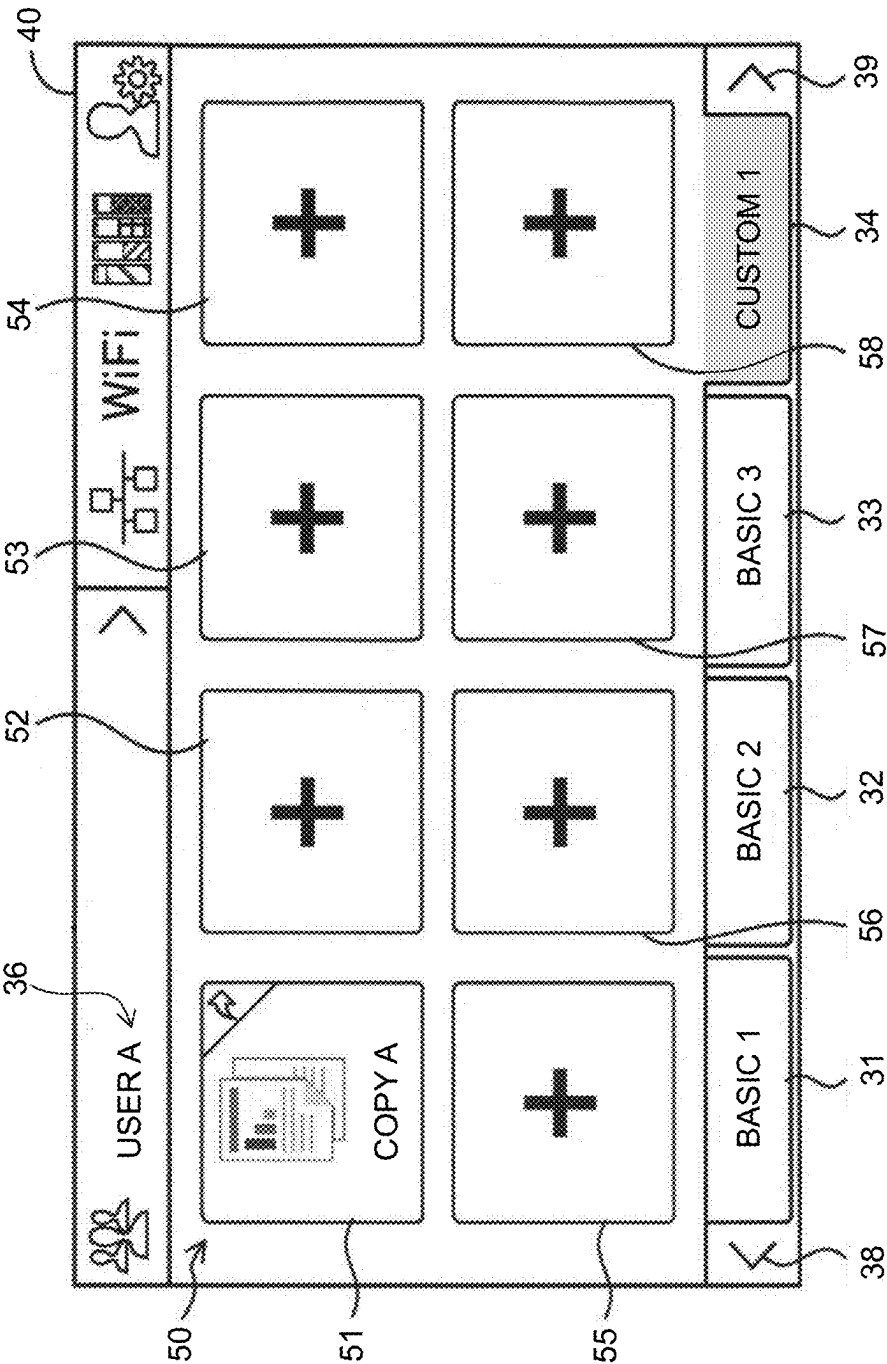


FIG. 6B

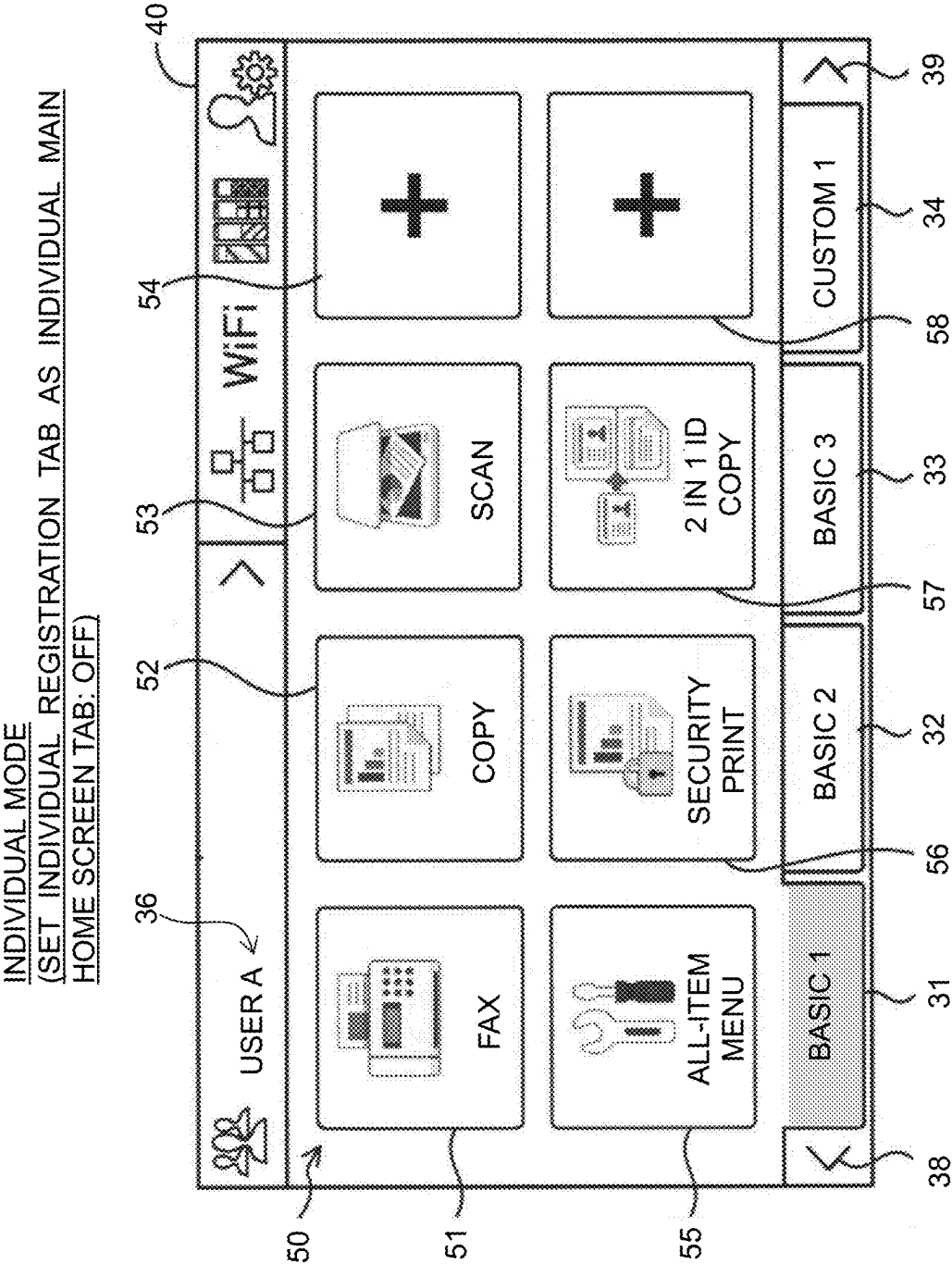


FIG. 6C

INDIVIDUAL MODE
(SET INDIVIDUAL REGISTRATION TAB AS INDIVIDUAL MAIN
HOME SCREEN TAB: OFF)
(FIRST TAB CHANGED TO PERSONAL ATTRIBUTE)

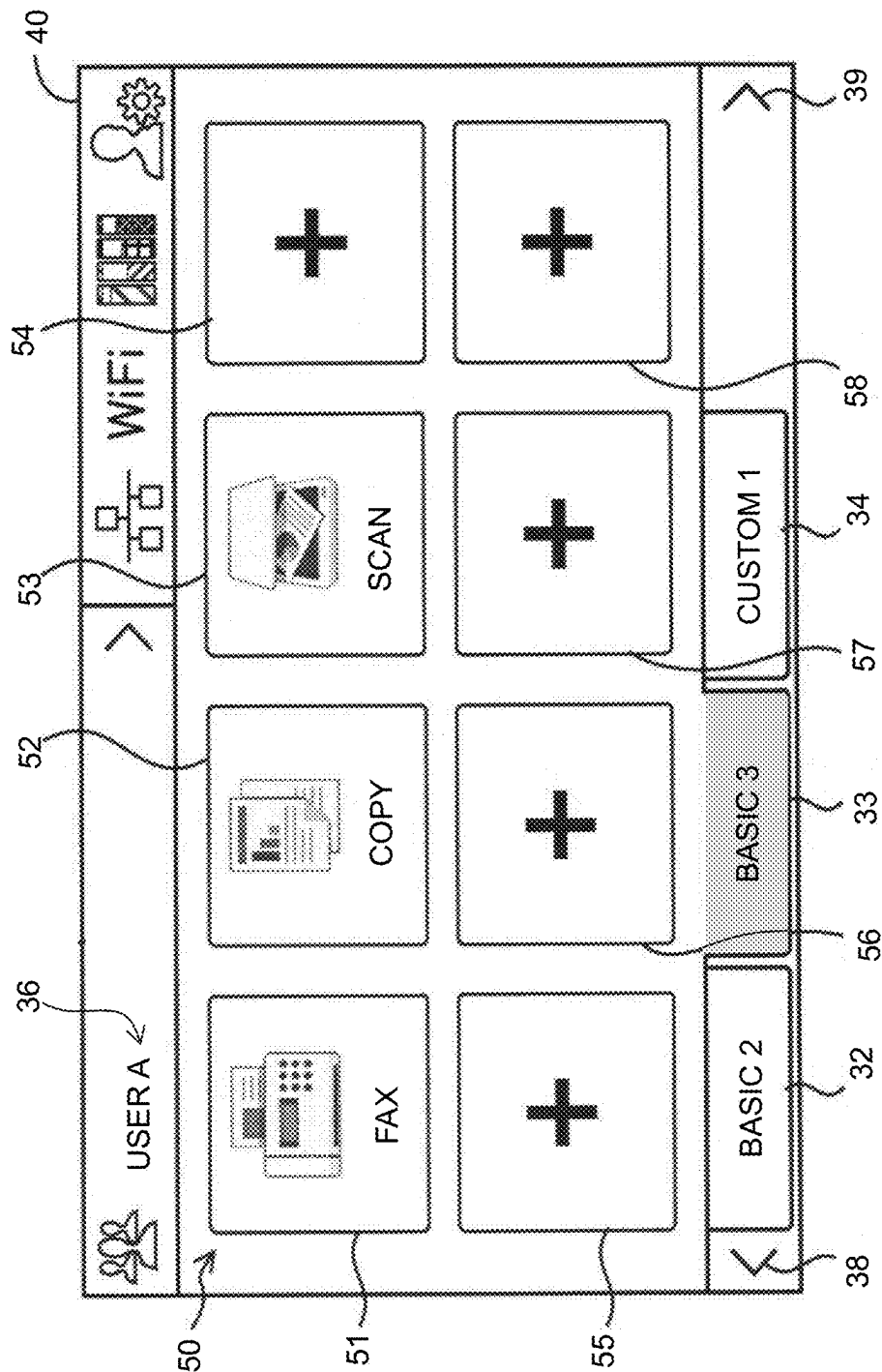


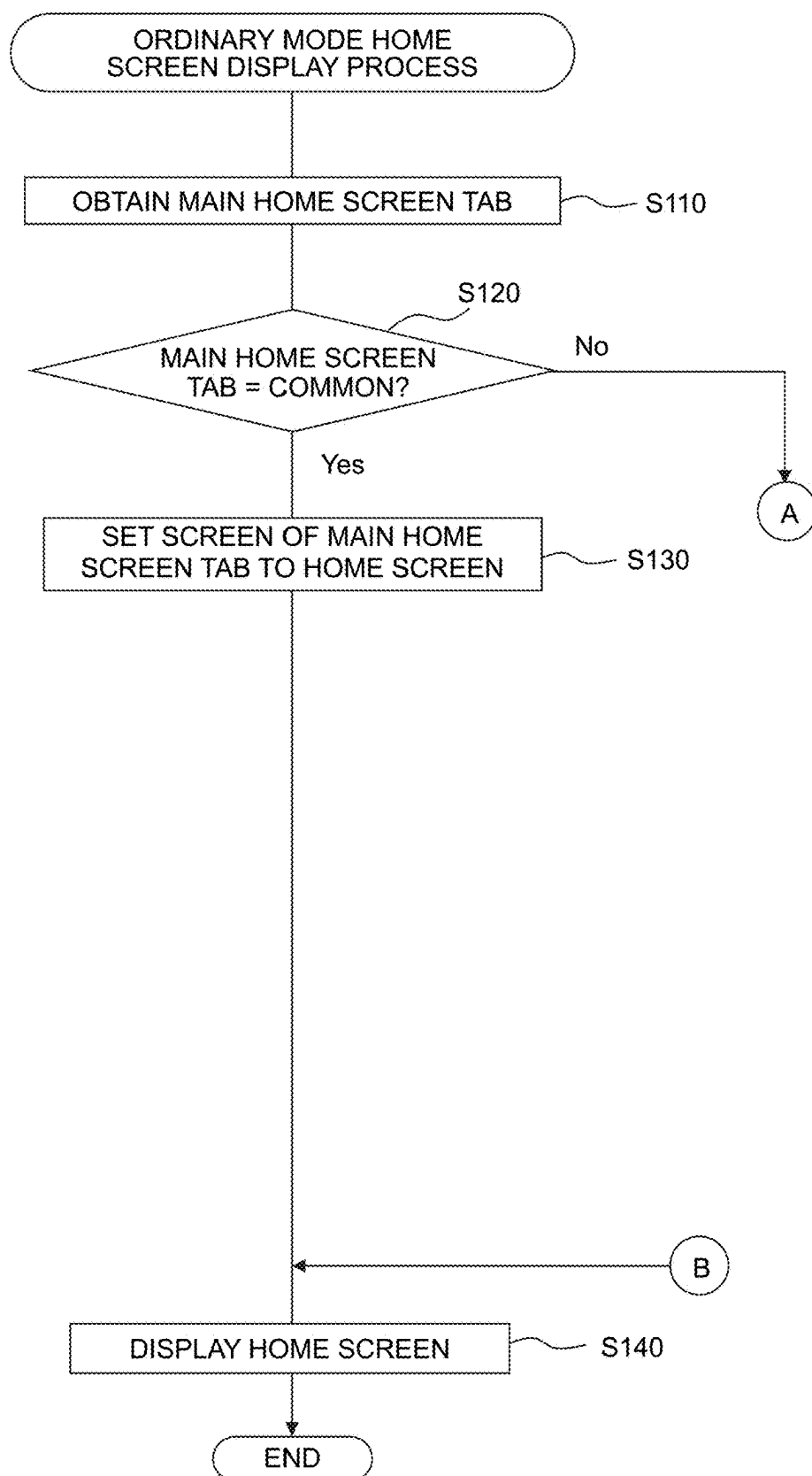
FIG. 7A

FIG. 7B

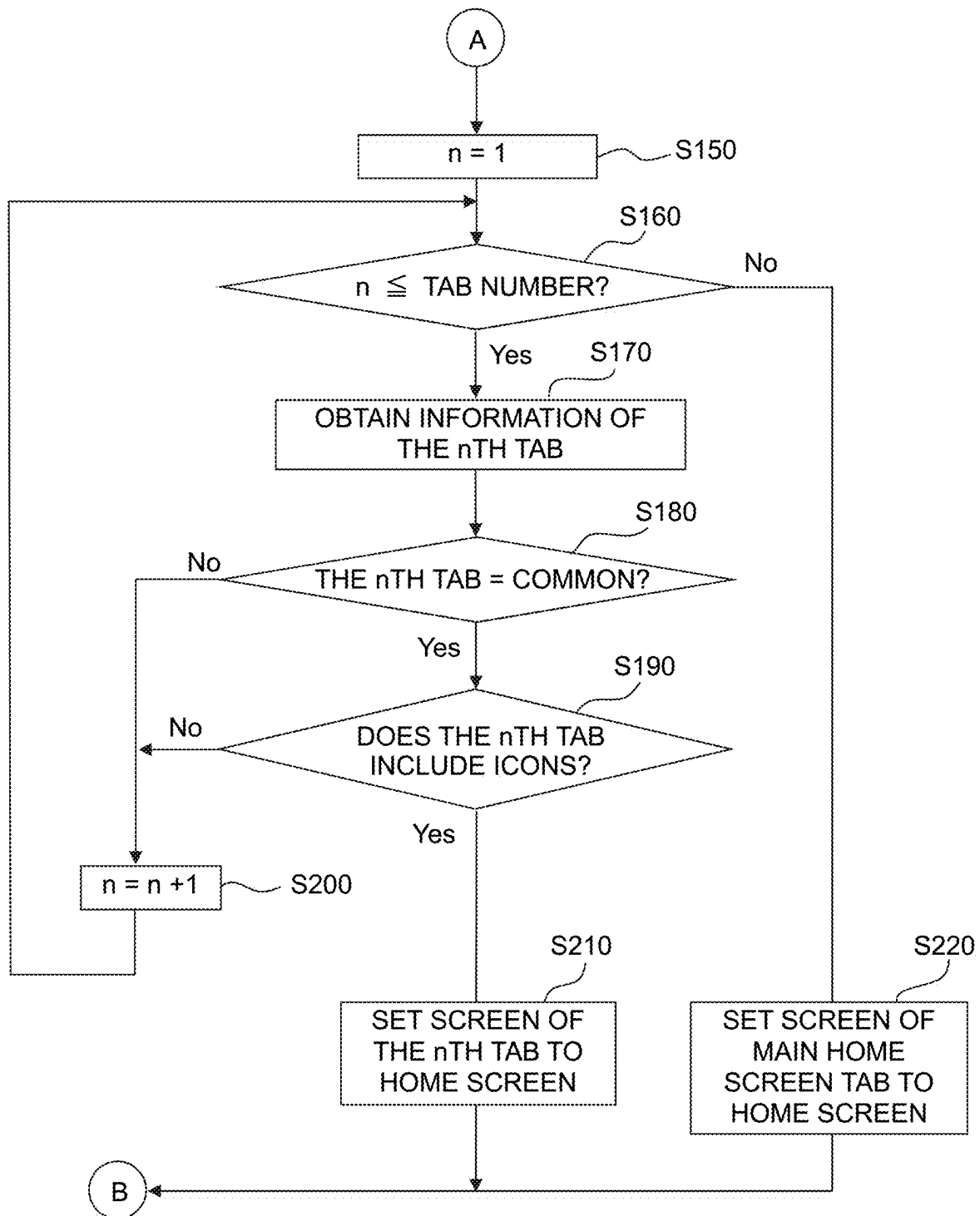


FIG. 8

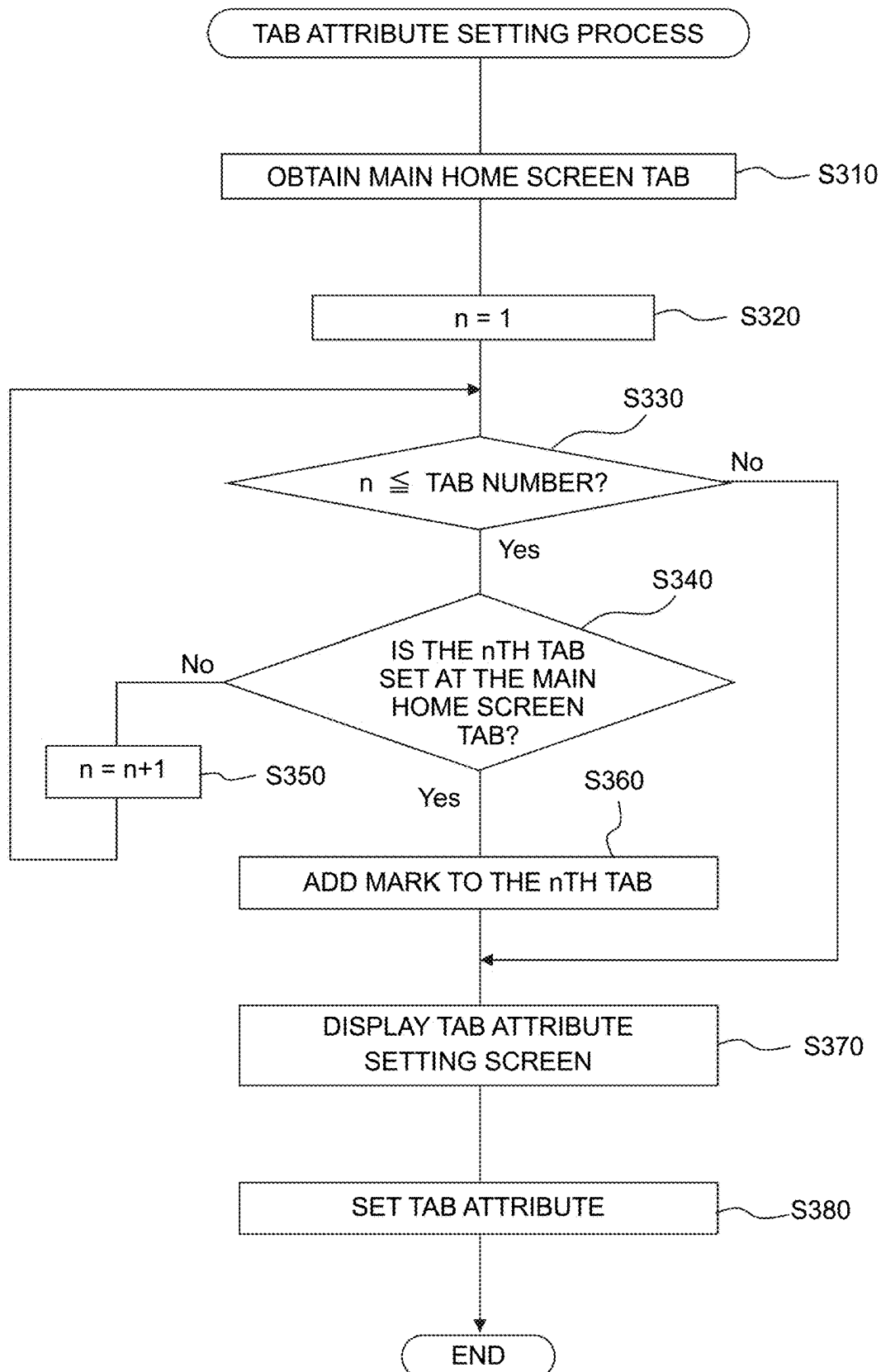


FIG. 9

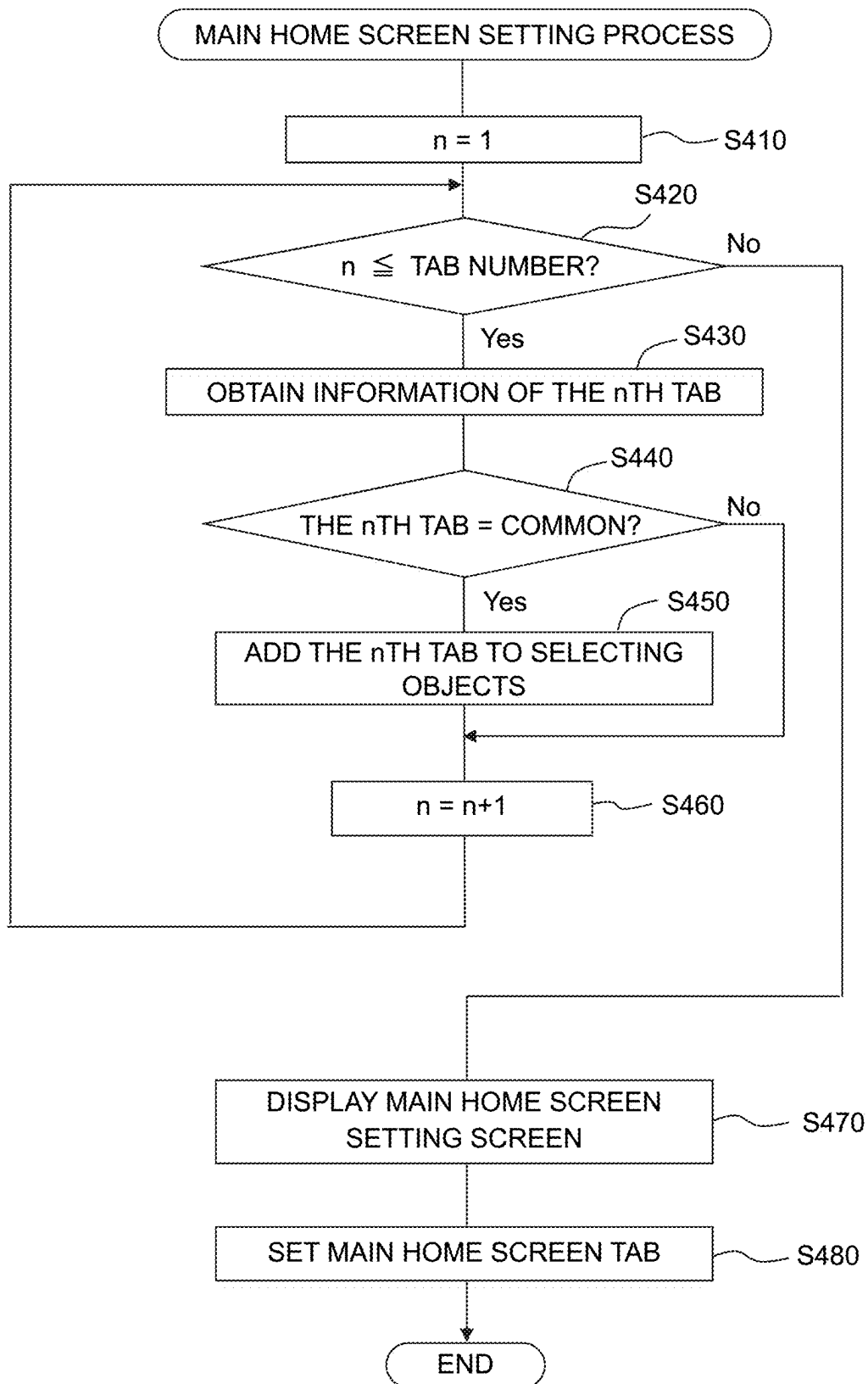


FIG. 10A

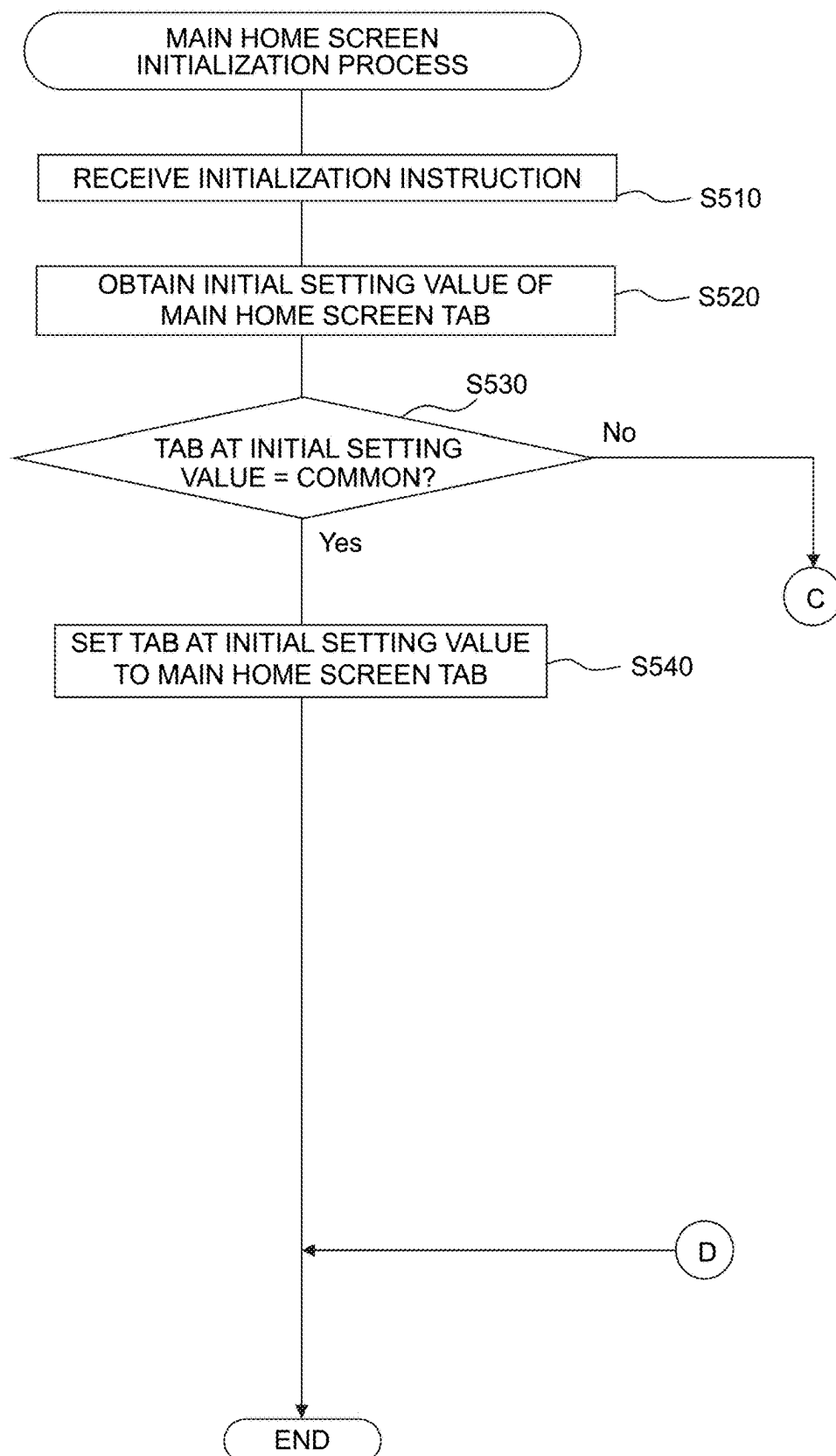


FIG. 10B

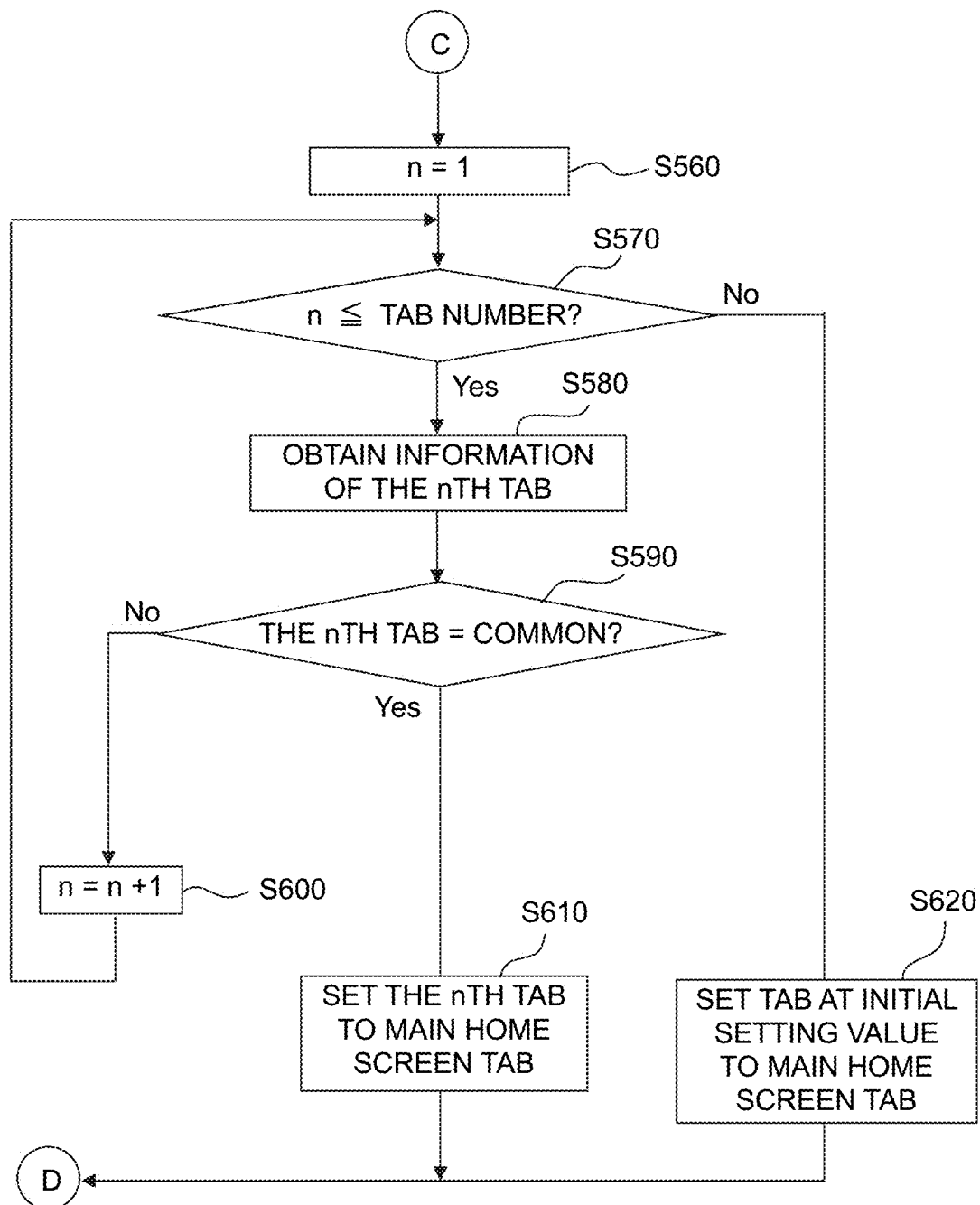


FIG. 11A

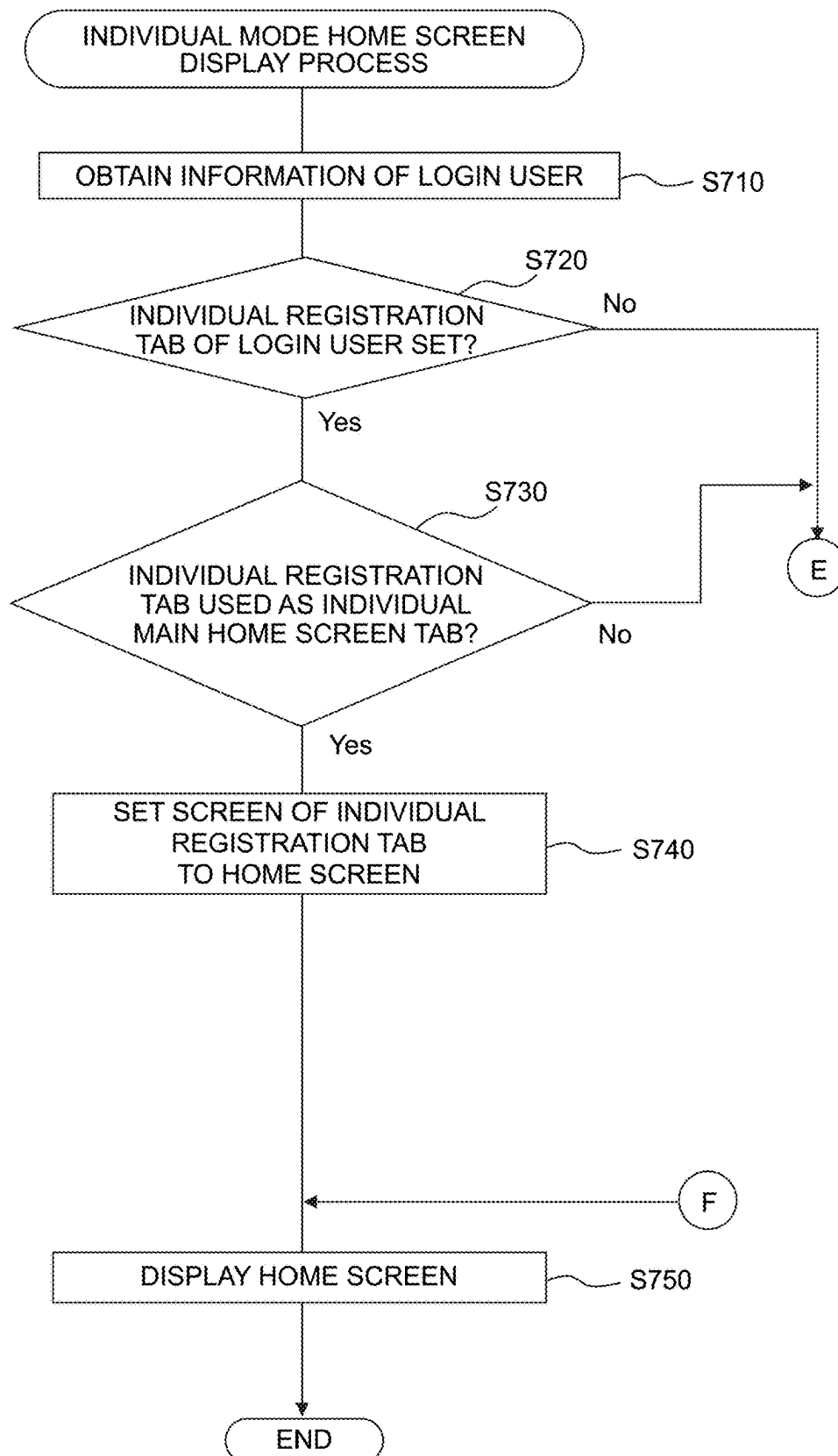


FIG. 11B

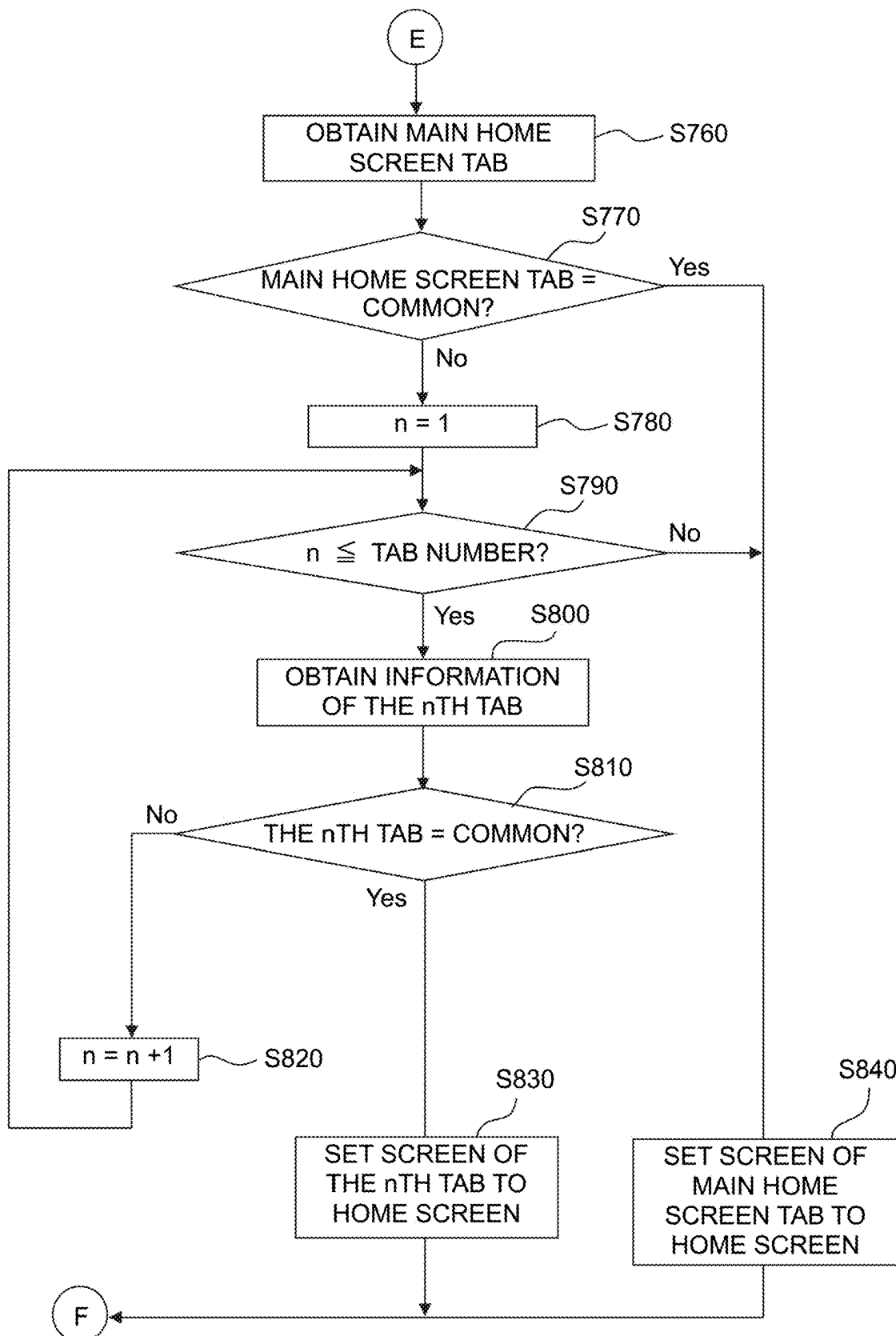


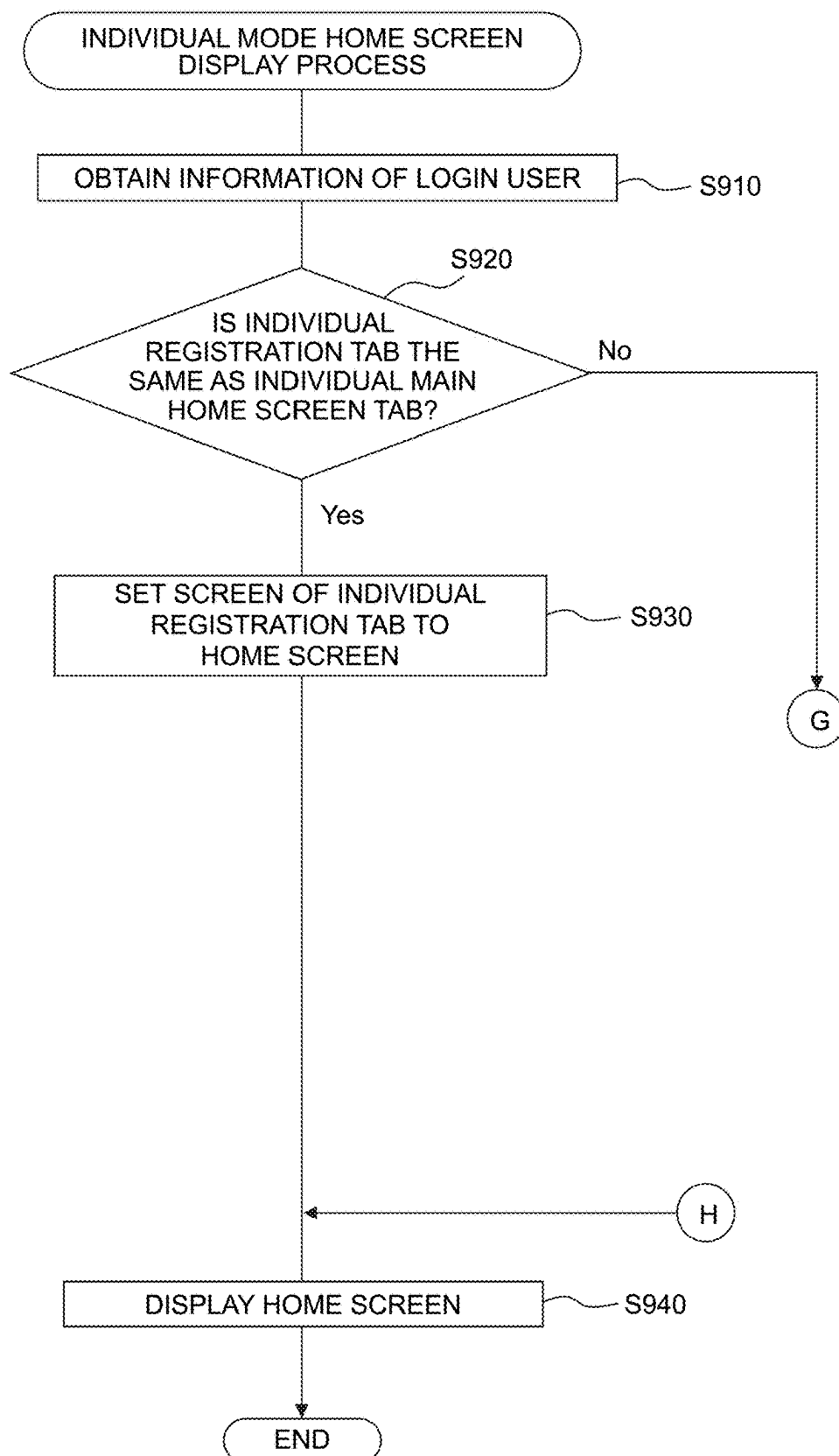
FIG. 12A

FIG. 12B

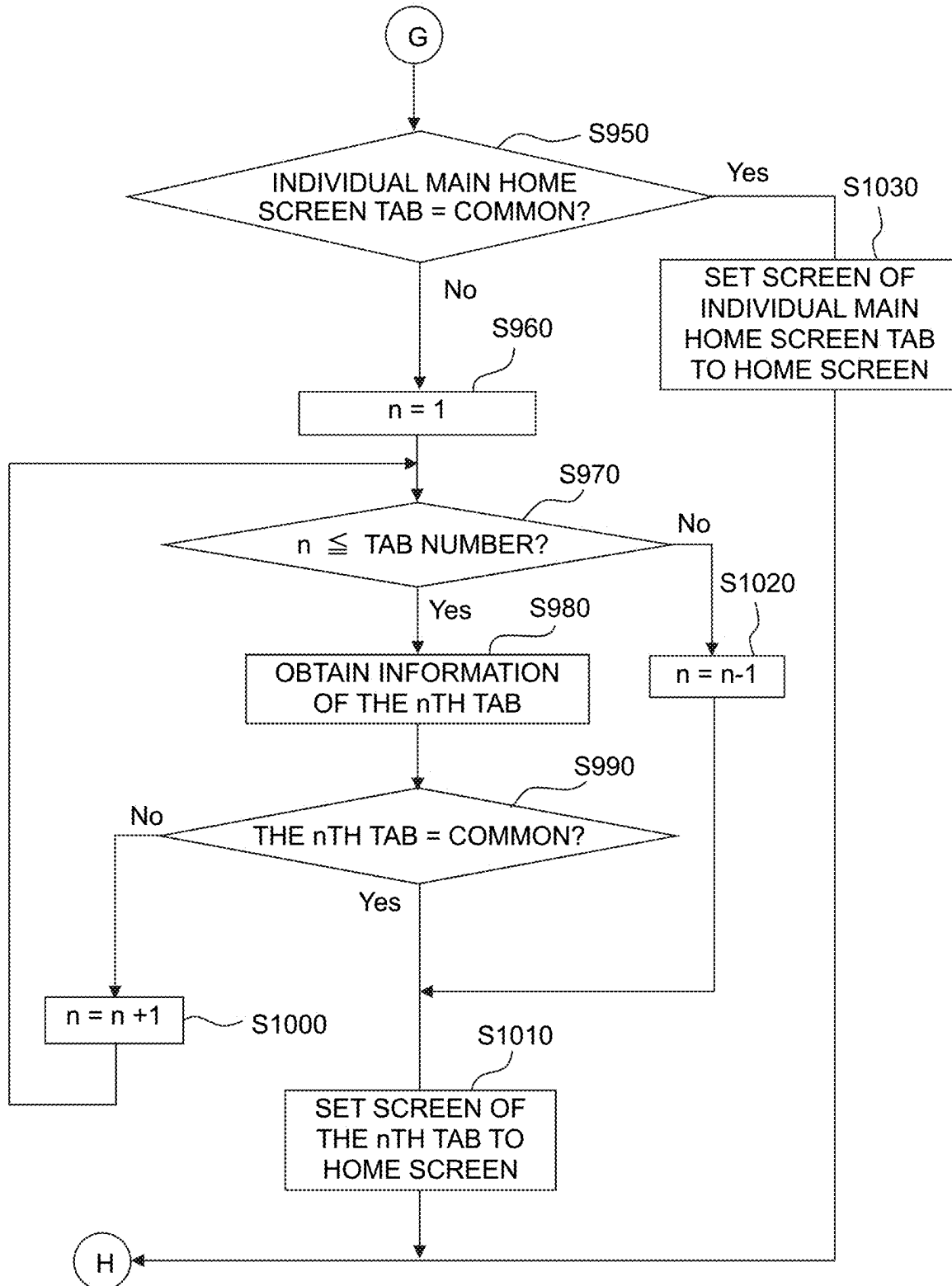


IMAGE PROCESSING APPARATUS**REFERENCE TO RELATED APPLICATIONS**

This Application is a Continuation of U.S. application Ser. No. 17/385,352 filed on Jul. 26, 2021, which is a Continuation of U.S. application Ser. No. 16/581,072 filed on Sep. 24, 2019 (now U.S. Pat. No. 11,082,572), which application claims priority from Japanese Patent Application No. 2018-184739, filed on Sep. 28, 2018, the disclosures of which are incorporated herein by reference in their entirety.

BACKGROUND ART

Japanese Patent Application Laid-open No. 2016-71680 discloses a multifunction apparatus where a shortcut list screen is displayed in a tab form. In the shortcut list screen of the tab form, if any one of the displayed tabs is set into a selected state, then such a screen is displayed as including options for a shortcut function associated with the tab in the selected state.

In the multifunction apparatus disclosed in Japanese Patent Application Laid-open No. 2016-71680, in an ordinary mode where no registered user has logged in, on the shortcut list screen, there is displayed a common tab which can be used by all users. On the other hand, in an individual mode where any registered user has logged in, on the shortcut list screen, the common tab is not displayed but only an individual tab is displayed for the registered user who has logged in.

SUMMARY

With respect to the multifunction apparatus disclosed in Japanese Patent Application Laid-open No. 2016-71680, such a usage pattern is conceivable, for example, as to preset any common tab for a main tab which is the initial value of the tab being first set into the selected state after starting the operation in the ordinary mode such as at the time of startup, at the time of switching to the ordinary mode, or the like.

However, problems as follows may occur with such kind of usage pattern. That is, in the case of such a configuration that it is possible to change the setting of attribute to the common tab or the individual tab for each tab, the main tab can be changed in attribute from the common tab to the individual tab. If the main tab is changed in attribute to the individual tab, then it is difficult for the multifunction apparatus to specify the tab which should be set into the selected state after starting the operation in the ordinary mode.

The present teaching is made in view of the above problems, and an object thereof is to provide a display control apparatus where a screen of tab form is displayed and, in the case of such a configuration that it is possible to determine the initial value of a tab being first set into the selected state after starting the operation in the ordinary mode, even if the attribute of the tab whose initial value is determined is changed to the attribute displayed in the individual mode, an appropriate screen is still displayed after the operation is started in the ordinary mode.

According to a first aspect of the present teaching, there is provided an image processing apparatus, including: a printer; a user interface; and a controller. In a case that the controller receives an input of authentication information of a user and that the user has logged in to the image processing apparatus based on the authentication information received by the controller, the controller is configured to: cause the

user interface to display one screen associated with the user logged in to the image processing apparatus, the one screen being one of a plurality of ordered screens and being neither a top screen nor a last screen of the plurality of ordered screens, the one screen being displayed on the user interface in a state of being switchable to a previous screen previous to the one screen and a next screen next to the one screen; in a case of receiving an operation by the user on the one screen to switch the one screen to the previous screen, cause the user interface to display the previous screen; and in a case of receiving an operation by the user on the one screen to switch the one screen to the next screen, cause the user interface to display the next screen.

According to a second aspect of the present teaching, there is provided an image processing apparatus, including: a printer; a user interface; and a controller. In a case that the controller receives an input of authentication information of a user and that the user has logged in to the image processing apparatus based on the authentication information received by the controller, the controller is configured to cause the user interface to display a main screen, the main screen being selected from a plurality of screens each of which is associated with at least a login user logged in to the image processing apparatus.

According to a third aspect of the present teaching, there is provided an image processing apparatus, including: a printer; a scanner; a display; an input interface; and a controller. The controller is configured to execute a display process for causing the display to display a standby screen. The standby screen is included in a plurality of screens and configured to be switchable to another screen. Icons, which are images associated with functions respectively, are registered for each of the screens. One of a common attribute and a personal attribute is set to each of the screens. A specific registered user is registered for each of the screens. The icons include at least one icon associated with a printing function of the printer and at least one icon associated with a scanning function of the scanner. The controller is further configured to execute: a main screen setting process for setting one of the screens as a main screen; and a user mode setting process for setting a user mode for the image processing apparatus to one of an ordinary mode in which no registered user has logged in to the image processing apparatus and an individual mode in which a registered user has logged in to the image processing apparatus, and for setting the user mode to the individual mode in accordance with an input of user authentication information corresponding to the registered user via the input interface. The display process includes: a first display process for causing the display to display the standby screen in accordance with the user mode being set to the ordinary mode; and a second display process for causing the display to display the standby screen in accordance with the user mode being set to the individual mode. In a case that the controller executes the first display process, a common attribute screen set with the common attribute is displayed as the standby screen, but no personal attribute screen set with the personal attribute is displayed in the display. If the main screen is set with the common attribute, then the main screen is displayed as the standby screen. If the main screen is set with the personal attribute, then the main screen is not displayed and another common attribute screen is displayed as the standby screen. In a case that the controller receives the input of the authentication information of the registered user and that the registered user has logged in to the image processing apparatus based on the authentication information received by the controller, the controller executes the second display

process, and in a case that the controller executes the second display process, a personal attribute screen registered with the registered user is displayed as the standby screen, and no personal attribute screen registered with another registered user is displayed as the standby screen.

According to the first to third aspects of the present teaching, if the user mode is set in the ordinary mode, then the personal attribute tab is not displayed and the common attribute tabs are displayed, and one of the common attribute tabs set as the main tab is set in the selected state. However, if the main tab is set with the personal attribute, then the main tab is not displayed and another common attribute tab is set in the selected state.

Therefore, even if the main tab is changed to the personal attribute, after the operation is started in the ordinary mode, it is still possible to appropriately determine a tab in the selected state, and thus appropriately display the standby screen.

BRIEF DESCRIPTION OF DRAWINGS

FIG. 1 is an illustration diagram depicting a schematic configuration of an information processing system.

FIGS. 2A and 2B are illustration diagrams depicting examples of a home screen in an ordinary mode in the case of setting a main home screen setting value to a first tab, wherein FIG. 2A shows an example of the home screen in the case where the first tab to a third tab are of a common attribute, and FIG. 2B shows an example of the home screen in the case where the first tab is changed from the state of FIG. 2A to a personal attribute.

FIG. 3A is an illustration diagram depicting a main home screen tab setting table, FIG. 3B is an illustration diagram depicting a tab setting table, FIG. 3C is an illustration diagram depicting a user list, and FIG. 3D is an illustration diagram depicting an individual main home screen tab setting table of an individual registration tab.

FIG. 4 is an illustration diagram depicting a tab attribute setting screen.

FIG. 5 is an illustration diagram depicting a main home screen setting screen.

FIGS. 6A to 6C are illustration diagrams depicting examples of the home screen in an individual mode in the case of setting the main home screen setting value to the first tab, wherein FIG. 6A shows an example of the home screen in the case where the setting is ON for the individual registration tab to be an individual main home screen tab, FIG. 6B shows an example of the home screen in the case where the setting is OFF for the individual registration tab to be the individual main home screen tab, and FIG. 6C shows an example of the home screen in the case where the first tab is changed from the state of FIG. 6B to the personal attribute.

FIGS. 7A and 7B depict a flow chart of an ordinary mode home screen display process.

FIG. 8 is a flow chart depicting a tab attribute setting process.

FIG. 9 is a flow chart depicting a main home screen setting process.

FIGS. 10A and 10B depict a flow chart of a main home screen initialization process.

FIGS. 11A and 11B depict a flow chart of an individual mode home screen display process.

FIGS. 12A and 12B depict a flow chart of another example of the individual mode home screen display process.

DESCRIPTION

Hereinbelow, referring to the accompanied drawings, an embodiment of the present teaching will be explained.

Outline of an Information Processing System

An information processing system 1 depicted in FIG. 1 according to this embodiment includes an image processing apparatus 10 and an information processing apparatus 20. The image processing apparatus 10 and the information processing apparatus 20 are capable of mutual data communication via a network 3. The network 3 may be any kind of networks. The network 3 may be, for example, either a wired network or a wireless network. Further, for the network either wired or wireless, the communication method in particular may be of any kind.

Configuration of the Image Processing Apparatus

The image processing apparatus 10 includes a number of functions such as a scanning function, a printing function, a copy function, and a facsimile function. The scanning function is a function to scan an image of a document sheet and generate image data of the scanned image. The printing function is a function to print an image on recording paper. The copy function is a function to print the image scanned with the scanning function, on a recording sheet with the printing function. The facsimile function is a function to communicate facsimile data.

As depicted in FIG. 1, the image processing apparatus 10 includes a controller 11, a memory 12, a display 13, an input interface 14, a scanner 15, a printer 16, a media interface 17, a first communication interface 18, and a second communication interface 19. These units of interface are connected with each other via a bus 5. Note that the input interface 14, the media interface 17, the first communication interface 18, and the second communication interface 19 will be abbreviated below as “input I/F 14”, “media I/F 17”, “first communication I/F 18”, and “second communication I/F 19”, respectively.

The controller 11 has a CPU. The memory 12 has semiconductor memories such as, for example, a ROM, a RAM, an NVRAM, a flash memory, and the like. That is, the image processing apparatus 10 is provided with a microcomputer including a CPU and semiconductor memories.

The controller 11 realizes various functions by executing programs stored in a non-transition physical recording medium. In this embodiment, the memory 12 plays the role of the non-transition physical recording medium in which the programs are stored. Note that the various functions realized by the controller 11 are not only realized by program execution but may be realized by using one or a number of hardware devices to replace part or all of the programs.

The memory 12 stores the data of various screens including an after-mentioned home screen 40 (see FIGS. 2A and 2B and FIGS. 6A to 6C). Further, the memory 12 also stores the data of a main home screen tab setting table depicted in FIG. 3A, the data of a tab setting table depicted in FIG. 3B, the data of a user list depicted in FIG. 3C, the data of an individual main home screen tab setting table depicted in FIG. 3D, the data of a tab attribute setting screen depicted in FIG. 4, the data of a main home screen setting screen 35 depicted in FIG. 5, the programs of respective processes depicted in FIGS. 7 to 12, and the like.

5

The display **13** has a display device for displaying various kinds of information. The display **13** according to this embodiment has a display device **13a** as the display device. The input I/F **14** has an input device for receiving various input operations. The input I/F **14** according to this embodiment has a touch panel **14a** as the input device. The touch panel **14a** is arranged to overlap with an image display area for image display on the display device **13a** of the display **13**.

The touch panel **14a** can detect an instructing operation performed by an instructing body contacting with or approaching the image display area of the display **13**. That is, the touch panel **14a** is configured to be capable of outputting position information indicating an instructing position at which that instructing operation is performed, when the instructing body performs the instructing operation on the image display area of the display **13**. The touch panel **14a** of this embodiment is configured to output the position information continuously or periodically during the instructing body performing the instructing operation.

Note that the touch panel **14a** may be configured to be capable of the detection only by contact as the instructing operation or be configured to be capable of the detection only by approach or be configured to be capable of the detection by either contact or approach.

The controller **11** obtains the position information outputted from the touch panel **14a** and, based on the obtained position information, it is possible to detect whether or not there is an instructing operation by the instructing body, the instructing position if the instructing operation is performed, and at least one kind of specified operations by the instructing body if the instructing operation is performed.

The specified operations detectable with the controller **11** include, for example, a tap operation, a flick operation, a drag operation, and the like. The tap operation is for the instructing body to come away at the same position after the instructing operation is performed by the instructing body. It is conceivable to have the instructing body, which is capable of performing the instructing operation, in various particular forms; for example, in the form of a fingertip, in the form of a specific instructing device such as a stylus pen or the like.

The scanner **15** includes an image sensor adapted to read in images from manuscripts and generate image data for the read-in images. The printer **16** has a printing mechanism to carry out printing by ink jet technologies or electrophotographic technologies, being capable of printing an image on a sheet of the recording paper.

The media I/F **17** is an interface unit installed with various storage media such as USB flash memories and the like. The media I/F **17** controls the writing and reading of data into and out of the installed storage media.

The first communication I/F **18** is a communication interface for connecting the image processing apparatus **10** to the network **3**. The image processing apparatus **10** may execute data communication via the first communication I/F **18** in a wired or wireless manner with various information processing apparatuses such as personal computers, smartphones, tablet terminals, and the like. Further, the image processing apparatus **10** may be connected to the Internet via the first communication I/F **18** to execute data communication with various other servers, the information processing apparatuses and the like via the Internet.

The second communication I/F **19** is an interface for executing communication with external devices via a communication network **150**. The communication network **150** connected to the second communication I/F **19** is, for example, a public telephone network in this embodiment.

6

Basically, transceiving facsimile data in the facsimile function is executed via the second communication I/F **19**.

Configuration of the Information Processing Apparatus

The information processing apparatus **20** is an information processing terminal of any of the various types such as, for example, a personal computer, a tablet terminal, a smartphone, or the like. As depicted in FIG. **1**, the information processing apparatus **20** includes a controller **21**, a memory **22**, an input interface **23**, a display **24**, and a communication interface **25**. Note that the input interface **23** and the communication interface **25** will be abbreviated below as “input I/F **23**” and “communication I/F **25**”, respectively.

The controller **21** has a CPU for example. The memory **22** has, for example, semiconductor memories such as a ROM, a RAM, an NVRAM, a flash memory, and the like. That is, the information processing apparatus **20** according to this embodiment is provided with a microcomputer including a CPU and semiconductor memories. The memory **22** stores data and various kinds of software.

The display **24** has a display device capable of displaying an image such as a liquid crystal display, an organic EL display, or the like. The input I/F **23** has an input device for receiving various input operations. The input device of the input I/F **23** includes, for example, a keyboard, a mouse, a touch panel, and the like. The touch panel is arranged to overlap with the image display area on the display device of the display **24**.

The communication I/F **25** is a communication interface for executing communication with external devices via the network **3**. The information processing apparatus **20** can execute data communication with the image processing apparatus **10** via the communication I/F **25**.

Explanation on the Home Screen

With the image processing apparatus **10**, if the controller **11** is activated, then the display device **13a** displays the home screen **40** as depicted in FIG. **2A** or FIG. **2B**, for example. The home screen **40** has a number of tabs **31**, **32**, and so on.

The tabs displayed on the home screen **40** may differ depending on whether an after-mentioned user authentication function is turned ON or OFF. Further, even if the user authentication function is set at ON, the tabs may still differ according to an after-mentioned user mode or login user. FIGS. **2A** and **2B** show an example of the home screen **40** when the user authentication function is set at ON and the user mode is set at an ordinary mode.

Each of the tabs **31**, **32**, . . . is associated individually with a tab screen **50**. The home screen **40** displays the tab screen **50** corresponding to one tab set in a selected state among the tabs **31**, **32**, The selected tab is displayed in a different form from other tabs. FIG. **2A** shows an example where the first tab **31** named “Basic 1” is selected and the first tab **31** is displayed in the different form from the other tabs (such as in different color, different size, etc.). FIG. **2B** shows an example where the third tab **33** named “Basic 3” is selected and the third tab **33** is displayed in the different form from the other tabs.

The memory **12** stores various kinds of information needed to display the home screen **40**, information about the

contents of the processes to be executed when various input operations are performed on the home screen **40** via the input I/F **14**, and the like.

A number of icons are arranged in the tab screen **50** associated with each tab. In this embodiment, for example, eight icons are arranged in one tab screen **50**. That is, as depicted in FIG. 2A, a first icon **51** to an eighth icon **58** are arranged in the tab screen **50**. It is possible to change the type of each icon.

Each icon arranged in the tab screen **50** is associated with a specific process in a specific function. The user can tap any of the icons to execute the specific process associated with that icon.

In the tab screen **50** exemplified in FIG. 2A, for example, the first icon **51** is associated with the specific process of the facsimile function. The second icon **52** is associated with the specific process of the copy function. The third icon **53** is associated with the specific process of the scanning function. The fourth icon **54** and the eighth icon **58** are unregistered icons and unassociated with any specific process. The user can register a specific process to any of those unregistered icons. Explanation is omitted for the fifth icon **55** to the seventh icon **57**.

The tabs displayed on the home screen **40** are controlled in the tab setting table depicted in FIG. 3B. N tabs (N is an arbitrary nonzero natural number) are listed up in the tab setting table according to a predetermined order. The predetermined order in this embodiment is, as depicted in FIG. 3B for example, the first tab, the second tab, the third tab, . . . , the N-th tab.

An attribute is set for each tab. In this embodiment, common or personal attribute is set. It is possible to change the tab attribute arbitrarily for each tab. FIG. 3B shows an example where the first tab to the third tab are of the common attribute while the other tabs are of the personal attribute.

Further, a tab name is set for each tab. The tab name is displayed in each tab when the tab is displayed on the home screen **40**. It is possible to change the tab name arbitrarily for each tab. Further, to each tab, there are set the number of associated icons or, more specifically, the number of the icons other than the unregistered icons.

The tab setting table may be prestored in the memory **12** of the image processing apparatus **10**. In such a case, for the image processing apparatus **10**, the user may be allowed to arbitrarily customize the tab setting table stored in the memory **12**.

Further, the tab setting table may be obtained at a specific acquisition time from another device such as the information processing apparatus **20** or the like, and then be stored in the memory **12**. In this embodiment, for example, in the information processing apparatus **20**, the tab setting table is configured to be allowed for arbitrary customization. Then, if the controller **11** of the image processing apparatus **10** is activated, then the tab setting table is obtained via the network **3** from the information processing apparatus **20**, and then stored in the memory **12**. If the tab setting table is stored in the memory **12**, then the controller **11** refers to the stored tab setting table and then displays the home screen **40**.

Note that after being stored in the memory **12**, the tab setting table already stored in the memory **12** may still be updated by a new tab setting table which is obtained from the information processing apparatus **20** at a specific time.

Each tab on the home screen **40** has the rightward direction as its arrayal direction in this embodiment. That is, each tab is arrayed in order in the rightward direction. The arrayal order of each tab along the arrayal direction is, for

example, an order according to the aforementioned predetermined order, that is, an ascending order according to the predetermined order.

In this embodiment, it is possible to display four tabs, for example, at the most at the same time on the home screen **40**. In this embodiment, the total number N of tabs is more than four which is the maximum number of tabs displayable on the home screen **40** at the same time (to be referred to below as "the maximum number of simultaneously displayable tabs"). FIG. 2A shows an example where the first tab **31** to the third tab **33** are displayed on the home screen **40** among the N tabs from the first tab to the N-th tab.

In the vicinity of the two ends of the area displaying the tabs on the home screen **40**, a left button **38** and a right button **39** are displayed, respectively. If the tabs to be displayed are five or more and four of the same are displayed, then it is possible to display the not displayed tab(s) by tapping the left button **38** or the right button **39** to left-scroll or right-scroll the tabs to be displayed.

If the user desires to switch the tab screen with the currently selected tab to the tab screen with another tab for example, then he or she only needs to tap the tab of the switch-to tab screen. If a tab other than the currently selected tabs is tapped, then that tapped tab is selected, and the tab screen corresponding to that newly selected tab is displayed on the home screen **40**.

The image processing apparatus **10** is provided with the user authentication function which is set to ON or OFF. Setting the user authentication function to ON or OFF may be performed by the image processing apparatus **10** or be performed via the network **3** by an external device such as the information processing apparatus **20** or the like.

As depicted in FIGS. 2A and 2B, if the user authentication function is set at ON, then the user authentication function is enabled, and the user mode of the image processing apparatus **10** is set to the ordinary mode or an individual mode. The ordinary mode is a user mode in which any registered user has not logged in. That is, it is possible for all users to set the ordinary mode without needing an authentication process such as inputting a password or the like. On the other hand, the individual mode is to be set by any preregistered user (to be referred to below as "registered user") who performs a predetermined user authentication operation to log in. That is, with any registered user logging in, the individual mode is set in accordance with that registered user.

In this embodiment, it is possible to register a user on the user list in the image processing apparatus **10** directly or indirectly via an external device such as the information processing apparatus **20** or the like. For example, the user list depicted in FIG. 3C is under the management of the information processing apparatus **20**. The registered users are registered on the user list. Further, it is possible to register an individual registration tab in the user list for each registered user. If the information processing apparatus **20** executes a registration process for a registered user, then that registered user is added to the user list. Further, it is possible to arbitrarily register an individual registration tab for that registered user.

The user list depicted in FIG. 3C exemplifies that at least the users A, B, C, and D are registered as the registered users. Further, there is depicted an example where the fourth tab is registered as the individual registration tab of the user A, the fifth tab is registered as the individual registration tab of the user B, the second tab is registered as the individual registration tab of the user C, and no tab is registered as the individual registration tab of the user D. It is also possible

to register a tab of either the common or personal attribute for the individual registration tab. In this embodiment, however, as one example, it is possible for one registered user to register only one tab such that it is not possible for a number of users to set the same one tab to the individual registration tab repetitively.

The user list may be managed by the image processing apparatus 10 per se as well but, basically, is to be managed by the information processing apparatus 20 and the image processing apparatus 10 may obtain the same from the information processing apparatus 20 as necessary. In this embodiment, each time a registered user logs in, the image processing apparatus 10 carries out a data communication with the information processing apparatus 20 to refer to the user list managed in the information processing apparatus 20 for obtaining the information indicating the individual registration tab corresponding to the registered user who has logged in.

The login of a registered user is performed by executing the predetermined authentication process. In the predetermined authentication process, for example, the registered user-specific authentication information (the password, for example) is inputted via the input I/F 14.

Depending on whether the user mode is the ordinary mode or the individual mode, the tabs displayed on the home screen 40 differ. Further, in the individual mode, too, depending on which registered user has logged in, the tabs displayed on the home screen 40 differ.

If the user authentication function is set to ON, then as depicted in FIGS. 2A and 2B, a user switching interface 36 is displayed on the home screen 40. The user switching interface 36 is an image for the user's operation (for example, a tap operation) in order to switch the user mode.

In the user switching interface 36, information is displayed to show the current user mode. In this embodiment, if the user mode is set at the ordinary mode, then as depicted in FIGS. 2A and 2B, the word "Public" is displayed to indicate the ordinary mode. If the user mode is set at the individual mode, then as depicted in FIGS. 6A to 6C, information is displayed to show the login user in the user switching interface 36.

If the user switching interface 36 is tapped, then the display 13 displays an undepicted user mode selection screen for selecting the user mode. On the user mode selection screen, the ordinary mode and each registered user are displayed as the options. Then, if the ordinary mode is selected, then the user mode is set to the ordinary mode.

In the ordinary mode, on the home screen 40, the tabs of the common attribute are displayed and one of those tabs is set to the selected state. In the ordinary mode, basically, no tabs of the personal attribute are displayed.

Note that if the user authentication function is set to ON, then when the controller 11 is powered and activated (to be referred to below as "first timing"), the controller 11 sets the user mode to the ordinary mode for example, and displays the home screen 40 corresponding to the ordinary mode. Further, in the ordinary mode, if a screen different from the home screen 40 is displayed, then when an undepicted home key included in the input I/F 14 is tapped (to be referred to below as "second timing"), the controller 11 also displays the home screen 40 corresponding to the ordinary mode. Further, if the user mode is changed from the individual mode to the ordinary mode (to be referred to below as "third timing"), the controller 11 still displays the home screen 40 corresponding to the ordinary mode.

The main home screen tab, which is first set to the selected state on the home screen 40 displayed at the first timing to

the third timing, is set in the main home screen tab setting table depicted in FIG. 3A. The user can perform a specific setting operation to change the main home screen tab. FIG. 3A shows an example where the main home screen tab is set at the first tab. At the first timing to the third timing, the controller 11 refers to the main home screen tab setting table to obtain the setting value of the main home screen tab. Then, the tab indicated by the obtained setting value is displayed on the home screen 40 as in the selected state.

In this embodiment, as depicted in FIG. 3B, the first tab to the third tab are of the common attribute. Therefore, on the home screen 40 in the ordinary mode, as exemplified in FIG. 2A, the first tab 31, the second tab 32 and the third tab 33 are displayed. Further, based on the setting value of the main home screen tab setting table depicted in FIG. 3A, the first tab 31 is selected, and the tab screen 50 for the first tab 31 is displayed.

The user can arbitrarily change the attribute and the name of each tab in the tab setting table. Therefore, such a case as follows may occur. That is, suppose that a certain user has set the first tab of the common attribute to the main home screen tab based on the setting value of the tab setting table depicted in FIG. 3B (corresponding to the state of FIG. 3A). At this point of time, at the first timing to the third timing, on the displayed home screen 40, as depicted in FIG. 2A, the first tab 31 to the third tab 33 of the common attribute are displayed while the first tab 31 set at the main home screen tab is set to the selected state.

Then, suppose that the first tab 31 is changed to the personal attribute. On this occasion, the first tab 31 is no longer displayed on the home screen 40 in the ordinary mode. On the other hand, the setting value for the main home screen tab is set to the first tab. Therefore, all tabs which ought to be in the selected state are not present in the home screen 40.

In this embodiment, therefore, in order to appropriately deal with the above case, if the tab set at the main home screen tab is changed to the personal attribute, then one of the tabs of the common attribute other than that tab is to be set to the selected state.

In the example of FIG. 3B, if the first tab is changed to the personal attribute, then there are two tabs of the common attribute: the second tab and the third tab. Any one of the two tabs may be set to the selected state but, in this embodiment, the tab at the upper position is selected according to the aforementioned predetermined order. Therefore, basically, the second tab is selected.

In this embodiment, however, the tabs without registered icons have a lower priority for the selected state than the tabs with registered icons. As depicted in FIG. 3B, if the numbers of icons of the second tab and the third tab are compared, then the second tab has no registered icon whereas the third tab has three registered icons.

Therefore, in this embodiment, considering the predetermined order, the second tab is arranged at the upper position over the third tab; however, the standard for determining whether or not icons are registered takes a priority. As a result, displayed on the home screen 40 as exemplified in FIG. 2B, the second tab 32 without registered icon is not in the selected state but the third tab 33 with the registered icons is in the selected state.

Next, using FIGS. 4 and 5, an explanation will be made on a method for changing the tab attribute and a method for changing the setting of the main home screen tab. The tab attribute can be changed via the tab attribute setting screen 30 depicted in FIG. 4, either at the image processing apparatus 10 or by remote control from the information

11

processing apparatus 20. The tab attribute setting screen 30 includes attribute setting images 71 to 78 according to each tab. The user can change the tab attribute via the attribute setting image corresponding to the tab whose attribute is to be changed among the attribute setting images 71 to 78.

Further, on the tab attribute setting screen 30, as depicted in FIG. 4, in the vicinity of the attribute setting image of the tab set at the main home screen tab (at the right side, for example), a main home setting mark 7 is added. Based on the main home setting mark 7, the user can know which tab is set at the main home screen tab. Therefore, it is possible to prevent the tab set at the main home screen tab from being changed to the personal attribute.

The setting value for the main home screen tab depicted in FIG. 3A can be set via a main home screen setting screen 35 of FIG. 5 displayed in the image processing apparatus 10. The main home screen setting screen 35 displays the option for each tab set in the tab setting table, that is, tab selection buttons 35a, 35b, and so on. The user can select any tab selection button to set the tab corresponding to that tab selection button to the main home screen tab.

However, on the main home screen setting screen 35, the tab selection buttons 35a, 35b, and 35c are effective as the options corresponding to the tabs of the common attribute, whereas the other tab selection buttons 35d, 35e, and so on are ineffective as the options corresponding to the tabs of the personal attribute. The tab selection buttons ineffective as the options are displayed in a different fashion (such as different color) from those tab selection buttons effective as the options. Even if the user performs a tap operation or the like, the operation will be ineffective.

On the other hand, if a tap operation is performed, for example, on any of the tab selection buttons 35a, 35b, and 35c effective as the options, then the tab corresponding to the tapped tab selection button is set to the main home screen tab. Then, according to the setting contents, the setting value of the main home screen tab setting table of FIG. 3A is updated.

On the aforementioned user mode selection screen (not depicted) displayed by tapping the user switching interface 36, if any registered user is selected, then an undepicted authentication screen is displayed for carrying out the authentication process. Via this authentication screen, if legitimate user authentication information corresponding to the selected registered user is inputted, then the authentication i.e. the login is successful, and the user mode is set to the individual mode for the selected registered user.

In the individual mode, from the user list (see FIG. 3C), an individual registration tab registered for the corresponding registered user is obtained, and that individual registration tab is displayed on the home screen 40. Further, in the individual mode, the tabs of the common attribute are also displayed on the home screen 40. In the individual mode, the tabs of the personal attribute are not displayed on the home screen 40 except the individual registration tab of the corresponding registered user.

In this embodiment, in the individual mode, to set which tab to the selected state is different with the setting value of the individual main home screen tab setting table of the individual registration tabs depicted in FIG. 3D (to be abbreviated below as "individual tab home setting table").

If the setting value of the individual tab home setting table is OFF, then one of the common attribute tabs to be displayed is set to the selected state (the upmost tab in the predetermined order, for example). On the other hand, if the setting value of the individual tab home setting table is ON,

12

then the individual registration tab corresponding to the login user is set to the selected state.

For example, the user A is on the user list where the fourth tab is registered as the individual registration tab. Therefore, if the user A has logged in, and the setting value of the individual tab home setting table is set at ON for example, then on the home screen 40, as depicted in FIG. 6A, the fourth tab 34 being the individual registration tab of the user A and the first tab 31 to the third tab 33 of the common attribute are displayed. Then, the fourth tab 34 is in the selected state.

Further, from the state depicted in FIG. 6A, for example, if the setting value of the individual tab home setting table is changed to OFF, then as exemplified in FIG. 6B, the first tab 31 is set to the selected state. Further, from the state depicted in FIG. 6B, for example, if the first tab 31 is changed to the personal attribute, then as depicted in FIG. 6C, the first tab 31 is deleted. Then, among the other tabs of the common attribute, the third tab 33 is set to the selected state, the third tab 33 being the upmost tab in the predetermined order and having the registered icons.

Ordinary Mode Home Screen Display Process

Next, referring to FIGS. 7A and 7B, an explanation will be made on an ordinary mode home screen display process executed in the ordinary mode for displaying the home screen 40. At any of the aforementioned first timing to third timing comes, the controller 11 reads in the program from the memory 12 and executes the same for the ordinary mode home screen display process.

If the ordinary mode home screen display process is started, then the controller 11 obtains the setting value of the main home screen tab from the main home screen tab setting table (see FIG. 3A) in the step S110. Further, referring to the tab setting table (see FIG. 3B), the controller 11 obtains the tab attribute, the tab name and the number of icons of each tab registered in the tab setting table. Note that the controller 11 may obtain information of the tab setting table from another device such as the information processing apparatus 20 or the like, and obtain the tab attribute from the tab setting table while storing the tab setting table into the memory 12. Alternatively, the controller 11 may read out the latest tab setting table obtained in the process of S110 and stored in the memory 12, and obtain the tab attribute, the tab name and the number of icons of each tab from the tab setting table.

In the step S120, the controller 11 determines whether or not the main home screen tab obtained in the step S110 is of the common attribute. That is, the controller 11 obtains the tab attribute matching the index in the obtained tab setting table from the setting value of the obtained main home screen tab, and then determines whether or not it is the common attribute. If the main home screen tab is of the common attribute, then the process proceeds to the step S130. In S130, the controller 11 sets the tab screen of the main home screen tab to the home screen. That is, the controller 11 sets the tab screen of the main home screen tab as the screen to be displayed on the home screen in the selected state. In the step S140, according to the contents set in S130, the home screen 40 is displayed. That is, referring to the tab setting table, the tab set to the home screen in the step S130 is set to the selected state on the home screen 40 including the tabs of the common attribute.

In the step S120, if the main home screen tab is not of the common attribute, then the process proceeds to the step S150. In S150, the variable n is set to one. In the step S160, the controller 11 determines whether or not the variable n is

13

equal to or less than the total number of tabs. If the variable *n* is equal to or less than the total number of tabs, then the process proceeds to the step S170. In S170, the controller 11 obtains information of the *n*th tab from the tab setting table.

In the step S180, based on the information obtained in the step S170, the controller 11 determines whether or not the *n*th tab is of the common attribute. If the *n*th tab is not of the common attribute, then in the step S200, the variable *n* is incremented by one, and then the process proceeds to the step S160. If the *n*th tab is of the common attribute, then the process proceeds to the step S190.

In the step S190, based on the information obtained in the step S170, the controller 11 determines whether or not the *n*th tab includes icons. If the *n*th tab does not include icons, then the process proceeds to the step S120. If the *n*th tab includes icons, then the process proceeds to the step S210. In S210, the tab screen of the *n*th tab is set to the home screen. After the step S210, the process proceeds to the step S140, and the home screen 40 is displayed according to the contents set in the step S210.

In the step S160, if the variable *n* exceeds the total number of tabs, then the process proceeds to the step S220. In S220, in the same manner as in the step S130, the tab screen of the main home screen tab is set to the home screen. After the step S220, the process proceeds to the step S140, and the home screen 40 is displayed according to the contents set in the step S220.

Tab Attribute Setting Process

Next, referring to FIG. 8, an explanation will be made on a tab attribute setting process for setting the tab attribute via the tab attribute setting screen 30 depicted in FIG. 4. If a request is inputted for setting a specific tab attribute, then the controller 11 of the image processing apparatus 10 reads in the program from the memory 12 and executes the same for the tab attribute setting process of FIG. 8. Note that the controller 21 of the information processing apparatus 20 may execute the tab attribute setting process and send the process result to the image processing apparatus 10 at a particular time.

If the tab attribute setting process is started, then in the step S310, the controller 11 or the controller 21 obtains the setting value of the main home screen tab from the main home screen tab setting table (see FIG. 3A).

In the step S320, the variable *n* is set to one. In the step S330, the controller 11 or 12 determines whether or not the variable *n* is equal to or less than the total number of tabs. If the variable *n* exceeds the total number of tabs, then the process proceeds to the step S370. If the variable *n* is equal to or less than the total number of tabs, then the process proceeds to the step S340.

In the step S340, the controller 11 or 12 refers to the main home screen tab setting table (see FIG. 3A), and determines whether or not the *n*th tab is set at the main home screen tab. If the *n*th tab is not set at the main home screen tab, then in the step S350, the variable *n* is incremented by one, and the process returns to the step S330.

If the *n*th tab is set at the main home screen tab, then the process proceeds to the step S360. In S360, the main home setting mark 7 (see FIG. 4) is set to be added to the *n*th tab. In the step S370, the tab attribute setting screen 30 (see FIG. 4) is displayed. On this occasion, the main home setting mark 7 is displayed in the vicinity of the attribute setting image of the tab set in the step S360 to add the main home setting mark 7. In the step S380, the user's input is received

14

at the tab attribute setting screen 30 displayed in the step S370, and the attribute of each tab is set based on the input.

Main Home Screen Setting Process

Next, referring to FIG. 9, an explanation will be made on a main home screen setting process for setting the main home screen tab (see FIG. 3A) via the main home screen setting screen 35 depicted in FIG. 5. If a request is inputted for setting a specific main home screen, then the controller 11 of the image processing apparatus 10 reads in the program from the memory 12 and executes the same for the main home screen setting process.

If the main home screen setting process is started, then the controller 11 sets the variable *n* to one in the step S410. In the step S420, the controller 11 determines whether or not the variable *n* is equal to or less than the total number of tabs. If the variable *n* is equal to or less than the total number of tabs, then the process proceeds to the step S430. In S430, the controller 11 obtains information of the *n*th tab from the tab setting table.

In the step S440, based on the information obtained in the step S430, the controller 11 determines whether or not the *n*th tab is of the common attribute. If the *n*th tab is not of the common attribute, then the variable *n* is incremented by one in the step S460 and then the process returns to the step S420. If the *n*th tab is of the common attribute, then the process proceeds to the step S450 to add the *n*th tab to selecting targets, and the process proceeds to the step S460.

In the step S420, if the variable *n* exceeds the total number of tabs, then the process proceeds to the step S470. In S470, the main home screen setting screen 35 (see FIG. 5) is displayed. On this occasion, as exemplified in FIG. 5, only such options are selectable as to correspond to the tabs added in the step S450 to the selecting targets (that is, the tabs of the common attribute), whereas such options are not selectable as to correspond to the tabs not added in the step S450 to the selecting targets (that is, the tabs of the personal attribute).

In the step S480, the user's input is received at the main home screen setting screen 35 displayed in the step S470, and the main home screen tab is set based on the input.

Main Home Screen Initialization Process

Next, referring to FIGS. 10A and 10B, an explanation will be made on the main home screen initialization process for initializing the setting value of the main home screen tab (see FIG. 3A). If a specific initialization instruction is inputted, then the controller 11 reads in the program from the memory 12 and executes the same for the main home screen initialization process of FIGS. 10A and 10B.

If the main home screen initialization process is started, then the controller 11 receives the initialization instruction operation in the step S510. In the step S520, the controller 11 obtains the initial setting value of the main home screen tab. The initial setting value of the main home screen tab may be obtained from anywhere. For example, it may be prestored in the memory 12 and then obtained from the memory 12. Alternatively, for example, it may be obtained from an external device such as the information processing apparatus 20 or the like.

In the step S530, the controller 11 determines whether or not the tab at the initial setting value obtained in the step S520 is of the common attribute. In particular, the controller 11 obtains the tab attribute corresponding to the tab at the initial setting value from the tab setting table stored in the

15

memory 12, and determines whether or not the tab is of the common attribute. If the tab at the initial setting value is of the common attribute, then the process proceeds to the step S540. In S540, the controller 11 sets the tab screen of the tab at the initial setting value to the home screen. That is, the controller 11 sets the tab screen of the tab at the initial as the screen displayed in the selected state on the home screen.

In the step S530, if the tab at the initial setting value is not of the common attribute, then the process proceeds to the step S560. In S560, the variable n is set to one. In the step S570, the controller 11 determines whether or not the variable n is equal to or less than the total number of tabs. If the variable n is equal to or less than the total number of tabs, then the process proceeds to the step S580. In S580, the controller 11 obtains information of the nth tab from the tab setting table.

In the step S590, based on the information obtained in the step S580, the controller 11 determines whether or not the nth tab is of the common attribute. If the nth tab is not of the common attribute, then the variable n is incremented by one in the step S600, and the process returns to the step S570. If the nth tab is of the common attribute, then the process proceeds to the step S610. In S610, the nth tab is set to the main home screen tab and the main home screen tab setting table is updated to the setting value.

In the step S570, if the variable n exceeds the total number of tabs, then the process proceeds to the step S620. In S620, the tab at the initial setting value is set to the main home screen tab and the main home screen tab setting table is updated to the setting value.

Individual Mode Home Screen Display Process

Next, referring to FIGS. 11A and 11B, an explanation will be made on an individual mode home screen display process to be executed in the individual mode for displaying the home screen 40. If the user mode is changed from the ordinary mode to the individual mode and if the login user is changed in the individual mode, then the controller 11 reads in the program from the memory 12 and executes the same for the individual mode home screen display process.

If the individual mode home screen display process is started, then the controller 11 obtains the login user's information in the step S710. In particular, from the user list (see FIG. 3C), the controller 11 obtains the information corresponding to the registered user having newly logged in.

In the step S720, the controller 11 determines whether or not the individual registration tab is set for the registered user having logged in. If the individual registration tab is set for the registered user having logged in, then the process proceeds to the step S730. In S730, the controller 11 determines whether or not the individual registration tab set for the registered user having logged in is set for the use as an individual main home screen tab. In particular, the controller 11 determines whether the setting value of the individual tab home setting table (see FIG. 3D) is ON or OFF. If the setting value of the individual tab home setting table is OFF, then the process proceeds to the step S760. If the setting value of the individual tab home setting table is ON, then the process proceeds to the step S740.

In the step S740, the tab screen of the individual registration tab is set to the home screen. That is, the tab screen of the individual registration tab is set as the screen displayed in the selected state on the home screen. In the step S750, based on the contents set in the step S740, the home screen 40 is displayed. That is, the controller 11 refers to the tab setting table and sets the tab set to the home screen in the

16

step S740 to the selected state, on the home screen 40 including the common attribute tabs and the individual registration tab of the login user.

In the step S720, if the individual registration tab is not set for the registered user having logged in, then the process proceeds to the step S760. In S760, the controller 11 obtains the setting value of the main home screen tab from the main home screen tab setting table (see FIG. 3A).

In the step S770, the controller 11 determines whether or not the main home screen tab obtained in the step S760 is of the common attribute. In particular, the controller 11 obtains the attribute of the tab corresponding to the setting value of the obtained main home screen tab from the tab setting table stored in the memory 12, and determines whether or not the attribute is common. If the main home screen tab is of the common attribute, then the process proceeds to the step S840. In S840, the controller 11 sets the tab screen of the main home screen tab to the home screen. On this occasion, in the step S750, based on the contents set in the step S840, the home screen 40 is displayed. That is, the controller 11 refers to the tab setting table and sets the tab set to the home screen in the step S840 to the selected state, on the home screen 40 including the common attribute tabs and the individual registration tab of the login user.

In the step S770, if the main home screen tab is not of the common attribute, then the process proceeds to the step S780. In S780, the variable n is set to one. In the step S790, the controller 11 determines whether or not the variable n is equal to or less than the total number of tabs. If the variable n is equal to or less than the total number of tabs, then the process proceeds to the step S800. In S800, the controller 11 obtains information of the nth tab from the tab setting table.

In the step S810, based on the information obtained in the step S800, the controller 11 determines whether or not the nth tab is of the common attribute. If the nth tab is not of the common attribute, then in the step S820, the variable n is incremented by one, and then the process returns to the step S790. If the nth tab is of the common attribute, then the process proceeds to the step S830.

In the step S830, the tab screen of the nth tab is set to the home screen. After the step S830, the process proceeds to the step S750, and the home screen 40 is displayed according to the contents set in the step S830. In the step S790, if the variable n exceeds the total number of tabs, then the process proceeds to the step S840.

Effects of the Embodiment

In the image processing apparatus 10 of this embodiment, if the user mode is set at the ordinary mode, then the personal attribute tabs are not displayed whereas the common attribute tabs are displayed, among which the tab set to the main tab is in the selected state. However, if the main tab is set at the personal attribute, then that main tab is not displayed but any other common attribute tab is in the selected state.

Therefore, even if the main tab is changed to the personal attribute, after the operation is started in the ordinary mode, it is still possible to appropriately determine the selected tab and display the home screen appropriately.

Note that in this embodiment, the memory 12 is an example of the first memory and the second memory. The home screen 40 is an example of the standby screen. Each of the icons 51 to 58 is an example of the functional object. The main home screen tab is an example of the main tab. The main home screen setting screen 35 is an example of the main tab selection screen. The main home setting mark 7 is

17

an example of the specific mark. The attribute setting images 71 to 78 are an example of the selection image. The table of FIG. 3D is an example of the individual mode information. The setting value of the individual main home screen tab on the user list of FIG. 3C is an example of the selection designation information.

Further, the processes of the steps S140, S210, S750 and S940 are an example of the display process, wherein specifically the processes of the steps S140 and S210 are an example of the first display process, whereas the processes of the steps S750 and S940 are an example of the second display process. The processes of the steps S470, S480, S540, S610 and S620 are an example of the main tab setting process, wherein the step S470 is an example of the main tab selection screen display process. The process of the step S370 is an example of the attribute setting screen display process. The process of the step S380 is an example of the attribute setting process. The process of the step S510 is an example of the initialization receiving process. The process of the step S530 is an example of the attribute determination process.

Modified Embodiments

While the embodiment of the present teaching was explained above, the present teaching is not limited to the above embodiment but can be executed in various modified forms.

With respect to displaying what kind of home screen at first when switched to the individual mode, that is, displaying which tab set in the selected state, the determination may be made in a different process from the individual mode home screen display process depicted in FIGS. 11A and 11B.

For example, as depicted by the user list of FIG. 3C, any tab for each registered user may be allowed for preregistration as the individual main home screen tab. Then, if the individual main home screen tab is set for the login user, then on the home screen, that individual main home screen tab may be displayed and set to the selected state.

In order to realize this, instead of the individual mode home screen display process of FIGS. 11A and 11B, for example, another individual mode home screen display process depicted in FIGS. 12A and 12B may be carried out to display the home screen. A specific explanation will be made on the individual mode home screen display process depicted in FIGS. 12A and 12B.

If the individual mode home screen display process is started, then in the step S910, the controller 11 obtains the login user's information in the same manner as in the step S710. In the step S920, based on the user list, the controller 11 determines whether or not the individual registration tab of the login user is the same as the individual main home screen tab of the login user.

If the individual registration tab of the login user is the same as the individual main home screen tab of the login user, then the process proceeds to the step S930. On this occasion, the process of S930 and the following process of S940 are the same as the processes of S740 and S750 of FIGS. 11A and 11B.

In the step S920, if the individual registration tab of the login user is different from the individual main home screen tab of the login user, then the process proceeds to the step S950. In S950, the controller 11 determines whether or not the individual main home screen tab is of the common attribute. In particular, the controller 11 obtains the tab attribute corresponding to the obtained individual main home screen tab from the tab setting table stored in the

18

memory 12, and then determines whether or not it is the common attribute. If the individual main home screen tab is of the common attribute, then the process proceeds to the step S1030. In S1030, the controller 11 sets the tab screen of the individual main home screen tab to the home screen, and then the process proceeds to the step S940. On this occasion, in the step S940, according to the contents set in S1030, the home screen 40 is displayed. That is, referring to the tab setting table, the tab set to the home screen in the step S1030 is set to the selected state on the home screen 40 including the tabs of the common attribute, and the individual registration tab of the login user.

In the step S950, if the individual main home screen tab is not of the common attribute, then the process proceeds to the step S960. In S960, the variable n is set to one. In the step S970, the controller 11 determines whether or not the variable n is equal to or less than the total number of tabs. If the variable n is equal to or less than the total number of tabs, then the process proceeds to the step S980. In S980, the controller 11 obtains information of the nth tab from the tab setting table.

In the step S990, based on the information obtained in the step S980, the controller 11 determines whether or not the nth tab is of the common attribute. If the nth tab is not of the common attribute, then in the step S1000, the variable n is incremented by one, and then the process returns to the step S970. If the nth tab is of the common attribute, then the process proceeds to the step S1010.

In the step S1010, the tab screen of the nth tab is set to the home screen. After the step S1010, the process proceeds to the step S940, and the home screen 40 is displayed according to the contents set in the step S1010. In the step S970, if the variable n exceeds the total number of tabs, then the process proceeds to the step S1020. In S1020, the variable n is decremented by one, and the process proceeds to the step S1010.

In the tab attribute setting screen 30 depicted in FIG. 4, the main home setting mark 7 may be designed in any form. Further, instead of adding the main home setting mark 7, it is possible to distinguish the corresponding tab from the other tabs by displaying that tab in the manner of, for example, emphasized presentation, changed color, or the like.

On the main home screen setting screen 35 depicted in FIG. 5, the options for the not selectable tabs may be displayed in any form. Alternatively, the options for the not selectable tabs may not be displayed at all.

Additionally, a number of functions possessed by one component in the above embodiment may be realized by a number of components, or one function possessed by one component may be realized by a number of components. Conversely, a number of functions possessed by a number of components may be realized by one component, or one function possessed by a number of components may be realized by one component. Further, part of the configuration in the above embodiment may be omitted. Further, at least part of the configuration of the above embodiment may be added to or may substitute the configuration of any of the above modified embodiments.

What is claimed is:

1. An image processing apparatus, comprising:

a printer;

a user interface; and

a controller configured to:

in a case that the controller receives an input of authentication information of a user and that the user has

19

logged in to the image processing apparatus based on the authentication information received by the controller,

cause the user interface to display one screen associated with the user logged in to the image processing apparatus, the one screen being one of a plurality of ordered screens and being neither a top screen nor a last screen of the plurality of ordered screens, the one screen being displayed on the user interface in a state of being switchable to a previous screen previous to the one screen and a next screen next to the one screen;

in a case of receiving an operation by the user on the one screen to switch the one screen to the previous screen, cause the user interface to display the previous screen; and

in a case of receiving an operation by the user on the one screen to switch the one screen to the next screen, cause the user interface to display the next screen.

2. The image processing apparatus according to claim 1, wherein the controller is configured to receive the input of the authentication information via a screen displayed on the user interface.

3. The image processing apparatus according to claim 1, wherein the controller is configured to cause the user interface to display a screen, which is associated with the user and registered as a main screen in a memory, as the one screen.

4. The image processing apparatus according to claim 3, wherein

the memory is configured to store a setting as to whether the screen, which is associated with the user and registered as the main screen in the memory, is displayed as the one screen, and

in a case of the memory storing the setting that the screen, which is associated with the user and registered as the main screen in the memory, is displayed as the one screen, the controller is configured to cause the user interface to display the screen, which is associated with the user and registered as the main screen in the memory, as the one screen.

5. An image processing apparatus, comprising:

a printer;

a user interface; and

a controller configured to:

in a case that the controller receives an input of authentication information of a user and that the user has logged in to the image processing apparatus based on the authentication information received by the controller,

cause the user interface to display a main screen, the main screen being selected from a plurality of screens each of which is associated with at least a login user logged in to the image processing apparatus.

6. The image processing apparatus according to claim 5, wherein the main screen is selected from a plurality of screens each of which is associated with the user and a plurality of screens each of which is associated with no user.

7. The image processing apparatus according to claim 6, wherein in a case of receiving an instruction to switch the main screen via the user interface in a state of the main screen being displayed in the user interface, the controller is configured to cause the user interface to switch the main screen to another screen which has not been selected as the main screen, the other screen being included in the group

20

consisting of the plurality of screens each of which is associated with the user and the plurality of screens each of which is associated with no user.

8. The image processing apparatus according to claim 7, wherein

in a case that no user has logged in to the image processing apparatus, the controller is configured to:

cause the user interface to display a second main screen, the second main screen being selected from a plurality of screens each of which is associated with no user; and

in a case of receiving an instruction to switch the second main screen via the user interface in a state of the second main screen being displayed in the user interface, cause the user interface to switch the second main screen to another screen which has not been selected as the second main screen, the other screen being included in the plurality of screens each of which is associated with no user.

9. An image processing apparatus, comprising:

a printer;

a scanner;

a display;

an input interface; and

a controller,

wherein the controller is configured to execute a display process for causing the display to display a standby screen, the standby screen being included in a plurality of screens and configured to be switchable to another screen,

wherein icons, which are images associated with functions respectively, are registered for each of the screens, wherein one of a common attribute and a personal attribute is set to each of the screens,

wherein a specific registered user is registered for each of the screens,

wherein the icons include at least one icon associated with a printing function of the printer and at least one icon associated with a scanning function of the scanner,

wherein the controller is further configured to execute:

a main screen setting process for setting one of the screens as a main screen; and

a user mode setting process for setting a user mode for the image processing apparatus to one of an ordinary mode in which no registered user has logged in to the image processing apparatus and an individual mode in which a registered user has logged in to the image processing apparatus, and for setting the user mode to the individual mode in accordance with an input of authentication information corresponding to the registered user via the input interface,

wherein the display process includes:

a first display process for causing the display to display the standby screen in accordance with the user mode being set to the ordinary mode; and

a second display process for causing the display to display the standby screen in accordance with the user mode being set to the individual mode,

wherein in a case that the controller executes the first display process,

a common attribute screen set with the common attribute is displayed as the standby screen, but no personal attribute screen set with the personal attribute is displayed in the display,

if the main screen is set with the common attribute, then the main screen is displayed as the standby screen, and

21

if the main screen is set with the personal attribute, then the main screen is not displayed and another common attribute screen is displayed as the standby screen, and wherein in a case that the controller receives the input of the authentication information of the registered user and that the registered user has logged in to the image processing apparatus based on the authentication information received by the controller, the controller executes the second display process, and

in a case that the controller executes the second display process,

a personal attribute screen registered with the registered user is displayed as the standby screen, and

no personal attribute screen registered with another registered user is displayed as the standby screen.

10. The image processing apparatus according to claim 9, wherein in the first display process, if the main screen is set with the personal attribute, then the controller is configured to display the other common attribute screen, which is other than the main screen and registered with the icons, as the standby screen.

11. The image processing apparatus according to claim 9, wherein

the main screen setting process includes a main screen selection screen display process for causing the display to display a main screen selection screen,

the main screen selection screen includes screens set with the common attribute as options, and does not include any screens set with the personal attribute as the options, and

in the main screen setting process, the controller is configured to set one screen, which is selected via the input interface among the screens displayed as the options on the main screen selection screen, as the main screen.

12. The image processing apparatus according to claim 9, wherein the controller is further configured to execute:

an attribute setting screen display process for causing the display to display an attribute setting screen including a selection image for selecting the attributes via the input interface for each of the screens, and for adding a specific mark to the selection image corresponding to the main screen; and

an attribute setting process for setting the attributes of the screens to one of the common attribute and the personal attribute selected via the input interface on the attribute setting screen.

13. The image processing apparatus according to claim 9, wherein

the controller is further configured to execute:

22

an initialization receiving process for receiving an initialization instructing operation via the input interface to set the standby screen displayed through the display process to a specific initial state; and

an attribute determination process for determining an attribute of a preset initial setting screen if the initialization instructing operation is received in the initialization receiving process, and

in the main screen setting process, the controller is configured to set the preset initial setting screen as the main screen in accordance with the preset initial setting screen being determined as having the common attribute in the attribute determination process, and to set the preset initial setting screen as the common attribute screen other than the main screen in accordance with the preset initial setting screen being determined as having the personal attribute in the attribute determination process.

14. The image processing apparatus according to claim 9, further comprising a first memory storing individual mode information,

wherein the individual mode information indicates which one is displayed as the standby screen between the main screen and the personal attribute screen registered with the registered user corresponding to the individual mode, in the standby screen displayed through the second display process in the individual mode, and

wherein in the second display process, the controller is configured to cause the display to display one of the main screen and the personal attribute screen registered with the registered user corresponding to the set individual mode as the standby screen, based on the individual mode information stored in the first memory.

15. The image processing apparatus according to claim 9, further comprising a second memory storing selection designation information,

wherein the selection designation information indicates a selection designation screen to be set as the standby screen in the second display process for each registered user, and

in the second display process, the controller is configured to cause the display to display the standby screen where the selection designation screen for the registered user corresponding to the set individual mode is set as the standby screen, based on the selection designation information stored in the second memory.

* * * * *